

# Cohesive Closure Repair and Its Hodge, Kernel, and Projection Shadows:

From Nonlinear Defects to Tangent Semigroups, with the Physical-Action Boundary

Peter Nero

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## Abstract

Modal Triplet Theory repeatedly uses fixed points, positive Hessians, heat kernels, coherent projectors, and perturbative graphs. These objects have often appeared as neighboring constructions without one theorem stating their exact common source. This paper supplies that theorem at the structural tier. Begin with a nonlinear defect section  $\Phi$  on a metric configuration space and the positive repair cost  $\mathcal{C} = \frac{1}{2} \|\Phi\|^2$ . Its negative gradient is closure repair. At a zero-defect fixed point the Hessian is exactly the normal operator  $D\Phi^\dagger D\Phi$ ; its heat semigroup, Green operator, spectral projectors, local kernel, and complete finite repair-vertex jet are then different mathematical shadows of the same linearized source. At a nonzero-defect critical point an additional covariant residual term is present, so the normal-square formula cannot be used without qualification.

The construction is extended to local Čech descent and to an  $\alpha$ -twisted cohesive module. The projective twist cancels in the endomorphism algebra, the Maurer–Cartan curvature  $F(a) = da + a^2$  gives a canonical nonlinear repair residual, and a chosen Hermitian structure yields the Hodge operator and tangent heat flow. An exact stratified example shows simultaneously that the full cohesive operator may remain regular while every uniform fiberwise Green operator diverges across a cohomology-rank jump. Compatible isometric intertwiners transport the repair flow, functional calculus, projectors, kernels, and formal graph coefficients.

The paper also proves the necessary boundary. A positive repair square is not a signed physical action. It loses first-order sign, phase, Morse index, causal support, and generally the physical interaction vertices. A direct action requires a variational anchor satisfying Helmholtz conditions; a universal multiplier lift always exists; and a cyclic Maurer–Cartan source admits a Chern–Simons-type signed action. For the q79 program, the cohesive Hartshorne–Serre benchmark closes the structural chain exactly after a Hermitian choice. Selection of the physical  $V_3/W_9$  endpoint, its HYM metric, full Hull–Strominger action, Lorentzian domains, and the continuum-to-finite intertwiner remain open. Thus closure repair explains why the operator and kernel language should recur, without promoting that recurrence into a completed theory of dynamics or quantization.

## Version 13 Revision Note

**Supersedes** The local Version 12 manuscript; no public release is implied.

**Reason** Auxiliary-pencil and global-gerbe results needed a coherent account without confusing their ranks with the physical eta9 source.

**Resolution** Integrates eight records, explaining family versus fiber triviality, local versus global monodromy and affine compression. Distinguishes integral quotient and vanishing lattices, and qualifies Neron-model applicability.

**Retained result** All fifty-six earlier integrations and the foundational Fixed Points series retain their stated scope.

**Remaining boundary** A rational characteristic is not its normalized integral lift. An auxiliary witness is not a selected physical path or an analytic gerbe zero.

## Version 12 Revision Note

**Supersedes** The local Version 11 manuscript; no public release is implied.

**Reason** The primitive repair target and its connection to the terminal resolution needed the eight existing PBW/apolar/socle source results.

**Resolution** Explains response typing, ordering-orbit normalization, tensor-natural retraction, complement-independent quotient descent and the explicit terminal-to-socle comparison, with examples of the lost choices.

**Retained result** All forty-eight prior source integrations and the foundational Fixed Points series retain their stated scope.

**Remaining boundary** A finite-field pairing is not a positive physical metric. Reconstructing quotient bases is not executing all actual source returns. These imports do not promote a selected physical endpoint.

## Version 11 Revision Note

**Supersedes** The local Version 10 manuscript; no public release is implied.

**Reason** The repair/action boundary needed the established anomaly-to-string constructor, Bott–Chern deformation lift and constrained moment-map roles.

**Resolution** Connects these three results, distinguishes holomorphic from unit-normalized volume forms, and retains the pseudo-Kähler qualification. An adjoint Lie-algebra inclusion is not an associative-algebra inclusion.

**Retained result** All forty-five earlier source integrations survive. The ambient six-row compiler is retained with its volume-form roles clarified; a constrained Hessian is not an ambient spectral intertwiner.

**Remaining boundary** A complex anomaly primitive does not select a real physical flux, a positive Calabi metric, a common endpoint or its action. The foundational Fixed Points series is unchanged.

## Version 10 Revision Note

**Supersedes** The local Version 9 manuscript; no public release is implied.

**Reason** The BV/topology boundary needed the already-available first-order matter and normalized spectral-localization results.

**Resolution** Explains graded index, chain projectors, mirror and exotic-rank tests, the ordinary graph-Prym obstruction, and the exact 85-by-82 evaluation criterion with its derived cohomology shift.

**Retained result** All forty-one earlier source imports and the foundational Fixed Points series retain their stated scopes.

**Remaining boundary** The selected line and its evaluation entries, harmonic normalization and physical lifting operator are not supplied by an auxiliary rank witness. Equivariant duality retains its canonical-line data.

## Version 9 Revision Note

**Supersedes** The local Version 8 manuscript; no public release is implied.

**Reason** The physical-action discussion needed the exact BV reduction mechanism and its relation to the already-integrated rational q79 topology.

**Resolution** Explains the cotangent retract, zero-section obstruction, normalized unit-orientation lift, sharp Euler lifting obstruction and the distinct mode/representation gates.

**Retained result** The earlier thirty-eight source imports keep their stated scopes. The foundational Fixed Points series is unchanged.

**Remaining boundary** A right inverse for a supplied action is not its selected physical origin, a mass-gap theorem or quantum BV transport.

## Version 8 Revision Note

**Supersedes** The local Version 7 manuscript; no public release is implied.

**Reason** Local nested transfer needed its global overlap and actual q79 rational-topology context.

**Resolution** Imports the curved projective-overlap theorem and the 96-to-88 Hirsch contraction, with examples explaining cyclic-pairing compensation, strict topology and nontrivial coefficient interactions.

**Retained result** Earlier charge, residual, projection and transfer results keep their stated scopes. Fixed Points I–VI are unchanged.

**Remaining boundary** Topological contraction and algebraic descent do not select a physical metric, endpoint, analytic reduction or 27-state map.

## Version 7 Revision Note

**Supersedes** The local Version 6 manuscript; no public release is implied.

**Reason** The inner-charge discussion and finite transfer results needed their nested Witten and noncentral coefficient constructions explained together.

**Resolution** Integrates six local records from outer nilpotence through restricted contraction to a positive curved coefficient algebra. Distinguishes the self-adjoint coefficient matrix from the full module superconnection, cyclic from positive pairings, and minimal from nonminimal transfer.

**Retained result** Previous residual, projection, first-jet, charge and finite-response results remain at their stated scopes. Fixed Points I–VI are unchanged.

**Remaining boundary** The finite tensor model does not select a physical factorization, global shared-line realization, analytic domain, action or value.

## Version 6 Revision Note

**Supersedes** The local Version 5 manuscript; no public release is implied.

**Reason** The common-source discussion had not yet integrated the single-charge, curved-charge, central-curvature and admissible-tangent results.

**Resolution** Explains six finite/local source records: shared central characters, nilpotent factorization, phase-dependent curved squares, complementary Jordan/Lie readouts, central identifiability and intrinsic stability on a constrained phase circle.

**Retained result** The previous residual, action, projection, finite transfer and first-jet qualifications remain. Fixed Points I–VI are unchanged.

**Remaining boundary** An algebraic charge is not an electric charge; phase dependence is not unique phase recovery; a small center does not select physical constants. The physical source, admissible stratum, domains and shared-line realization still require their own construction.

## Version 5 Revision Note

**Supersedes** The local Version 4 manuscript; no public release is implied.

**Reason** Compatible projection alone did not distinguish compression, inverse-weighted elimination and higher-product transfer, or explain the additional harmonic modes required by finite covariance completion.

**Resolution** Integrates eleven source records into one comparison: leakage, finite gauge descent, the 36- and 144-dimensional calculi, and the 48-dimensional response retract with its nontruncating higher products. Separates a harmonic quotient from a flat degeneration and an all-arity theorem from exhaustive operation tables.

**Retained result** The residual, action and projector-completeness corrections remain. No foundational Fixed Points theorem is changed.

**Remaining boundary** Finite covariance and higher operations do not select a physical HYM endpoint, continuum intertwiner, action or measured coupling.

## Version 4 Revision Note

**Supersedes** The local Version 3 manuscript; no public release is implied.

**Reason** An older continuum-to-finite contract bounded the whole ambient harmonic projector using only an intertwining residual on the retained space.

**Resolution** Gives an exact counterexample, separates retained from ambient spectral sectors, and states the completeness and isolated-cluster conditions needed for projector comparison.

**Retained result** Exact reducing intertwiners still transport every retained spectral projector. The repair, residual and finite numerical results are unchanged.

**Remaining boundary** No additional ambient zero modes, common analytic domains and perturbative spectral separation require their own certificates.

## Version 3 Revision Note

**Supersedes** The local Version 2 manuscript; no new public release is implied.

**Reason** The physical residual had only a schematic description, and the projection statement did not distinguish a unitary equivalence from an isometric embedding with transverse coupling.

**Resolution** Displays the six additional residuals, their derivatives and the corrected 25-block rank-102 support. States the extra adjoint condition for embedded Hessian transport, explains transported metrics, and organizes the physical endpoint contract into three structured source types.

**Retained result** The residual-Hessian, cohesive, descent, repair-jet and action-boundary theorems remain at their stated tiers. Previously established local and finite results are not reopened.

**Remaining boundary** A structural support mask is not a numerical matrix; symmetry reduction requires endpoint equivariance. Physical source selection, the common analytic domain and the four-dimensional action remain distinct.

## Version 2 Revision Note

**Supersedes** Version 1. The previous release remains public until a new release is approved.

**Reason** Connect strict-gradient repair to the finite-mode recurrence obstruction.

**Resolution** Connects strict repair with the stationary-ensemble and finite-mode recurrence obstructions.

**Retained result** Correctly scoped results and explanatory examples remain; no physical source-selection claim is promoted.

**Remaining boundary** Realization hypotheses and physical source or apparatus bridges remain separate obligations. This is an unreleased authoring revision.

## Version 1 revision note

**Supersedes.** No earlier paper. Version 1 establishes the baseline for this new theorem-owner paper.

**Reason.** The Fixed Points, wave-particle, and diagrammatics papers each use a downstream part of the same mathematical chain, but none is the proper owner of the upstream derivation from nonlinear closure repair.

**Resolution.** Version 1 proves and explains the structural chain

nonlinear closure defect  $\longrightarrow$  repair flow  $\longrightarrow$  linearized Hodge operator,  
linearized Hodge operator  $\longrightarrow$  kernel and projection shadows.

It consolidates separately certified theorem packets, proves their composition in standalone form, includes the nonzero-defect correction, and separates the positive repair functional from a signed physical action.

**Retained result.** The exact component theorems and their executable witnesses are retained at their established tiers. Their ownership is reorganized rather than weakened: this paper owns the composition, while the downstream papers retain their fixed-point, wave-particle, and graph theorems.

**Remaining boundary.** No observed mass, coupling, mixing angle, cosmological datum, or fitted coefficient enters the theorems. The paper does not prove a Lorentzian action, interacting QFT, ultraviolet-complete quantum gravity, unique actualization, or selection of the physical endpoint. The cohesive application remains an exact structural benchmark rather than a physical promotion.

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# 1 The question before the operator

Many physical formalisms begin with an operator. One specifies a Hamiltonian, a Laplacian, a Dirac operator, a Hessian, or a propagator and then asks what it does. MTT begins one question earlier:

What is being repaired, and why should its first observable shadow be an operator at all?

The answer developed here is deliberately modest. Suppose a state is admissible when a nonlinear defect vanishes. The squared size of the defect is then a natural nonnegative cost. Its gradient points in the direction in which inadmissibility grows most rapidly; its negative gradient therefore defines a repair flow. Linearizing that flow at an exact admissible state produces a positive normal operator. Once that operator exists, the heat semigroup, Green operator, spectral projectors, and local kernels belong to ordinary functional analysis.

This does not make every physical operator a theorem of MTT. It identifies one precise mechanism by which a large and important class of operators can arise. The mechanism has hypotheses, and several of them carry physical content: the configuration space, defect bundle, metric, connection, domain, background, and interpretation must all be selected. Hiding any one of these choices inside the word “closure” would merely relocate the input.

The central distinction of this paper is therefore:

**The repair operator is derived from a selected defect and metric. The selected defect and metric are not derived by notation.**

## 1.1 How to read the paper

Readers interested in the universal differential geometry may read [sections 2](#) and [3](#). The cohesive and q79 construction is in [sections 4](#) and [19](#). The operator, kernel, and projection chain is proved in [sections 6](#) and [7](#). The relation to perturbative graphs is in [section 13](#). The crucial warning that repair is not automatically action appears in [section 14](#). The final status and theorem-ownership tables are in [sections 21](#) and [23](#).

## 1.2 Vocabulary of claim strength

We use four labels throughout.

1. **Exact** means proved from the displayed assumptions.
2. **Exact structural benchmark** means proved on the named q79 object, but not promoted to the selected physical endpoint.
3. **Conditional transport** means the conclusion follows if the stated chain, metric, or transform intertwiner exists.
4. **Open physical promotion** means that the required source object or comparison has not been constructed and certified.

These labels are part of the mathematics. They prevent a theorem about an available representation from being mistaken for a theorem selecting the representation realized in nature.

## 2 Residual geometry and the repair Hessian

### 2.1 The nonlinear source

Let  $(\mathcal{M}, g)$  be a finite-dimensional Riemannian manifold. The same formulas extend to a Hilbert manifold when the derivatives, adjoints, and domains used below exist. Let

$$(\mathcal{E}, h, \nabla^\mathcal{E}) \longrightarrow \mathcal{M}$$

be a real metric or complex Hermitian vector bundle with metric connection, and let

$$\Phi \in \Gamma(\mathcal{E})$$

be a twice differentiable defect section. In the complex case every scalar pairing below means its real part. Define

$$\mathcal{C}(p) = \frac{1}{2}h(\Phi(p), \Phi(p)). \quad (2.1)$$

At  $p \in \mathcal{M}$ , write

$$r_p = \Phi(p), \quad (2.2)$$

$$J_p X = \nabla_X^\mathcal{E} \Phi, \quad (2.3)$$

$$B_p(X, Y) = \nabla_X^\mathcal{E} \nabla_Y^\mathcal{E} \Phi - \nabla_{\nabla_X^\mathcal{M} Y}^\mathcal{E} \Phi. \quad (2.4)$$

The tensor  $B_p$  is the covariant second derivative of the defect.

**Theorem 2.1** (Covariant residual-Hessian identity). *For every  $X, Y \in T_p \mathcal{M}$ ,*

$$d\mathcal{C}_p(X) = h(r_p, J_p X), \quad (2.5)$$

$$\text{Hess}_p \mathcal{C}(X, Y) = h(J_p X, J_p Y) + h(r_p, B_p(X, Y)). \quad (2.6)$$

*Equivalently,*

$$\text{grad } \mathcal{C}(p) = J_p^\dagger r_p, \quad H_p = J_p^\dagger J_p + S_{r_p}, \quad (2.7)$$

*where*

$$g(S_{r_p} X, Y) = h(r_p, B_p(X, Y)).$$

*The residual form is symmetric, including when  $\nabla^\mathcal{E}$  has curvature.*

*Proof.* Metric compatibility gives

$$X[\mathcal{C}] = h(\nabla_X^\mathcal{E} \Phi, \Phi),$$

which proves (2.5). Differentiate once more and subtract the Levi–Civita correction:

$$\begin{aligned} \text{Hess } \mathcal{C}(X, Y) &= X[d\mathcal{C}(Y)] - d\mathcal{C}(\nabla_X^\mathcal{M} Y) \\ &= h(\nabla_X^\mathcal{E} \Phi, \nabla_Y^\mathcal{E} \Phi) + h\left(\Phi, \nabla_X^\mathcal{E} \nabla_Y^\mathcal{E} \Phi - \nabla_{\nabla_X^\mathcal{M} Y}^\mathcal{E} \Phi\right). \end{aligned}$$

This is (2.6). The antisymmetric part of  $B$  is  $R^\mathcal{E}(X, Y)\Phi$ . Curvature of a metric or Hermitian connection is skew-adjoint, so  $\text{Re } h(\Phi, R^\mathcal{E}(X, Y)\Phi) = 0$ . The residual term is therefore symmetric.  $\square$

**Corollary 2.2** (Exact-zero normal square). *If  $p_*$  is an exact closure state,  $\Phi(p_*) = 0$ , then*

$$\text{grad } \mathcal{C}(p_*) = 0, \quad \text{Hess}_{p_*} \mathcal{C} = J_*^\dagger J_* \geq 0. \quad (2.8)$$

Moreover,

$$\ker \text{Hess}_{p_*} \mathcal{C} = \ker J_*.$$

This is the exact source of the positive repair Hessian used later. It is also the point at which a common shortcut must stop. Equation (2.8) is not the general Hessian formula.

**Corollary 2.3** (Nonzero-defect correction). *A state  $p$  is critical precisely when*

$$J_p^\dagger r_p = 0.$$

If  $r_p \neq 0$ , the Hessian is

$$J_p^\dagger J_p + S_{r_p},$$

and the second term generally cannot be reconstructed from  $r_p$  and  $J_p$  alone. On  $\ker J_p$ , local stability is controlled entirely by  $S_{r_p}$ .

**Example 2.4** (Why the correction cannot be dropped). Let  $\Phi_a : \mathbb{R} \rightarrow \mathbb{R}^2$  be

$$\Phi_a(x) = (x, x^2 + a), \quad a = 1.$$

At  $x = 0$ , the residual is  $(0, 1)$ , the derivative is  $(1, 0)$ , and  $J^\dagger r = 0$ . Thus  $x = 0$  is critical although the defect is nonzero. The Gram term is  $J^\dagger J = 1$ , the residual correction is 2, and

$$\text{Hess}_0 \mathcal{C} = 3.$$

Using only the normal square would give the wrong local stiffness by a factor of three.

## 2.2 Symmetry and physical zero modes

Let a Lie group  $G$  act isometrically on  $\mathcal{M}$  and fiberwise isometrically on  $\mathcal{E}$ , and suppose  $\Phi$  is equivariant.

**Proposition 2.5** (Orbit zero modes). *If  $p$  is critical and  $\xi_{\mathcal{M}}(p)$  is tangent to its  $G$ -orbit, then*

$$\text{Hess}_p \mathcal{C}(\xi_{\mathcal{M}}(p), Y) = 0$$

for every  $Y \in T_p \mathcal{M}$ .

*Proof.* Invariance gives  $d\mathcal{C}(\xi_{\mathcal{M}}) = 0$  identically. Differentiate in the direction  $Y$ . The term containing  $d\mathcal{C}(\nabla_Y \xi_{\mathcal{M}})$  vanishes at a critical point, leaving the claimed Hessian identity.  $\square$

Consequently, positivity in a gauge or phase-symmetric system means positivity on a slice normal to the symmetry orbit. Requiring strict positivity on the unreduced space would incorrectly reject the zero modes that encode the symmetry.

### 3 Repair as a flow

#### 3.1 Dissipative and reversible parts

Pure repair is the negative-gradient equation

$$\frac{du}{ds} = -\text{grad } \mathcal{C}(u). \quad (3.1)$$

The parameter  $s$  orders stabilization. It is not automatically Lorentzian time. A more general reversible–dissipative split is

$$\frac{du}{ds} = (K - \Gamma) \text{grad } \mathcal{C}(u), \quad (3.2)$$

where  $K$  is skew-adjoint and  $\Gamma$  is self-adjoint and positive semidefinite.

**Theorem 3.1** (Repair monotonicity and tangent generator). *Along every solution of (3.2),*

$$\frac{d}{ds} \mathcal{C}(u(s)) = -g(\text{grad } \mathcal{C}, \Gamma \text{grad } \mathcal{C}) \leq 0. \quad (3.3)$$

*At a critical point  $p$ , the linearized generator is*

$$D[(K - \Gamma) \text{grad } \mathcal{C}]_p = (K_p - \Gamma_p) H_p^\sharp. \quad (3.4)$$

*At an exact-zero state and for pure repair, the tangent equation is*

$$\partial_s v = -J_*^\dagger J_* v, \quad v(s) = e^{-s J_*^\dagger J_*} v(0). \quad (3.5)$$

*Proof.* Pair (3.2) with  $\text{grad } \mathcal{C}$ . The  $K$ -term vanishes by skew-adjointness, and positivity of  $\Gamma$  gives (3.3). At a critical point the derivatives of  $K$  and  $\Gamma$  multiply  $\text{grad } \mathcal{C}(p) = 0$ , so only the Hessian term remains. The exact-zero specialization follows from [theorem 2.2](#).  $\square$

**Corollary 3.2** (Strict-dissipation circulation no-go). *If  $\Gamma$  is positive definite along a periodic orbit of (3.2), that orbit is constant. A persistent circulation therefore requires a null-dissipation sector, a conservative sector, a relative equilibrium generated by symmetry, or external driving.*

This small result matters conceptually. A particle-like phase circulation cannot be obtained merely by drawing a loop around a strictly dissipative minimum. Repair explains stabilization. Reversible phase transport requires additional selected structure.

This also separates repair from stationary noise. Fixed Points III owns the stationary-ensemble obstruction for strict Lyapunov repair: under its integrability and invariance hypotheses, a stationary ensemble has zero dissipation almost everywhere. A closed finite-spatial-mode unitary model with compact resolvent supplies a different obstruction, namely recurrence rather than irreversible mixing. Removing only an occupation-number cutoff does not remove that finite-spatial-mode recurrence.

Thus a stationary diffusion limit needs an independently justified recurrent fast driver and a mixing/scaling argument, a continuum or thermodynamic limit, or a declared open system. Toral-chaos examples remain valid examples of separately supplied drivers; they are not flows of a strictly decreasing repair cost. These distinctions preserve the repair Hessian theorem and locate the extra dynamical input precisely. The owning discussion and certificates are in Fixed Points III and the [curated reproduction repository](#), under the finite-mode recurrence and fixed-point noise results; their proofs are not duplicated here.

### 3.2 Fixed points and basins

The flow may possess many fixed points or many attraction basins. Their existence, uniqueness, regularity, and convergence require the familiar compactness, coercivity, analytic-semigroup, or Łojasiewicz–Simon hypotheses developed in Fixed Points I [41]. The present paper does not duplicate those theorems. It supplies the upstream residual from which one eligible flow may be formed.

If a compact connected sector contains no zero of  $\Phi$ , the continuous cost  $\mathcal{C}$  attains a positive minimum. Such a minimum is a mathematically precise persistent-defect candidate. It is not automatically a particle: its physical interpretation still depends on sector selection, stability modulo symmetry, reversible transport, and the map to observables.

## 4 The cohesive source

### 4.1 Why a complex rather than only a bundle

The q79 source program contains an  $\alpha$ -twisted Hartshorne–Serre object

$$\mathcal{S}_{\text{HS}} \in D^b(J, \alpha)$$

with a bounded local perfect atlas. Its open-stratum contraction uses an inverse transverse coordinate and cannot pass through the exceptional stratum. Discarding the full complex would therefore erase precisely the place at which its cohomology rank changes.

Cohesive modules provide a lawful alternative. A bounded  $\alpha$ -twisted perfect atlas is represented by an  $\alpha$ -twisted graded smooth module  $E^\bullet$  with an integrable antiholomorphic superconnection

$$\overline{\mathbb{E}} = \overline{\mathbb{E}}_0 + \overline{\mathbb{E}}_1 + \overline{\mathbb{E}}_2 + \cdots, \quad \overline{\mathbb{E}}^2 = 0. \quad (4.1)$$

This is standard cohesive and twisted-complex mathematics [80, 3, 86, 87, 2, 5]. The q79-specific statement is that the certified local atlas satisfies its hypotheses.

**Theorem 4.1** (Endomorphism untwisting). *Let the local projective transitions of  $E^\bullet$  satisfy*

$$G_{ij}G_{jk}G_{ki} = \alpha_{ijk}\text{Id}.$$

*Then  $\text{End}(E^\bullet)$  glues as an ordinary global graded algebra by conjugation. The commutator differential*

$$d_{\text{End}}(T) = [\overline{\mathbb{E}}, T] \quad (4.2)$$

*satisfies  $d_{\text{End}}^2 = 0$ .*

*Proof.* On a triple overlap,

$$\text{ad}(G_{ij})\text{ad}(G_{jk})\text{ad}(G_{ki}) = \text{ad}(\alpha_{ijk}\text{Id}) = \text{Id}.$$

Thus the scalar gerbe cocycle disappears from the endomorphism transition. The graded Jacobi identity and  $\overline{\mathbb{E}}^2 = 0$  give

$$d_{\text{End}}^2(T) = [\overline{\mathbb{E}}^2, T] = 0.$$

□

The twist has not vanished from the object. It has become invisible to the adjoint algebra. This is exactly what is needed for a global deformation complex without falsely trivializing the original gerbe.

## 4.2 The canonical nonlinear residual

Set

$$\mathcal{A} = \Omega^{0,*}(\text{End } E), \quad d = [\mathbb{E}, -].$$

For a degree-one perturbation  $a \in \mathcal{A}^1$ , define

$$F(a) = da + a^2 = (\mathbb{E} + a)^2. \quad (4.3)$$

Then  $F(a) = 0$  precisely when the perturbed superconnection remains integrable. This is not an arbitrary polynomial chosen to resemble an interaction. It is the curvature of the perturbed cohesive source.

Choose a Hermitian structure and a gauge row. On degree one, let

$$\Phi_{\text{MC}}(a) = (F(a), d_0^\dagger a), \quad \mathcal{C}_{\text{MC}}(a) = \frac{1}{2} \|F(a)\|^2 + \frac{1}{2} \|d_0^\dagger a\|^2. \quad (4.4)$$

**Theorem 4.2** (Cohesive Maurer–Cartan repair). *At the integrable background  $a = 0$ ,*

$$D\Phi_{\text{MC}}(0) = (d_1, d_0^\dagger), \quad (4.5)$$

and

$$\text{Hess}_0 \mathcal{C}_{\text{MC}} = d_1^\dagger d_1 + d_0 d_0^\dagger = \Delta_1. \quad (4.6)$$

*The tangent negative-gradient flow is*

$$\partial_s a = -\Delta_1 a, \quad a(s) = e^{-s\Delta_1} a(0).$$

*The nonlinear curvature also satisfies the Bianchi identity*

$$d_a F(a) = 0, \quad d_a = d + [a, -].$$

*Proof.* Differentiate (4.3) at zero. The quadratic term has zero first derivative, so  $DF(0) = d_1$ . Apply [theorem 2.2](#) to the two residual rows in (4.4). The Bianchi identity follows by expanding  $dF + [a, F]$ , using  $d^2 = 0$  and the graded Leibniz rule.  $\square$

The algebraic residual is canonical once the cohesive object is fixed. The Hermitian metric, adjoint, normalization, and physical moment-map rows are not. In particular,  $d_0^\dagger a = 0$  is a gauge slice and must not be renamed the physical HYM equation.

## 5 One charge, complementary curvature readouts

The preceding construction starts from a defect and a metric. The next question is whether its reversible, dissipative and curvature operators can come from fewer independent choices. Six finite/local source records give a precise partial answer. Here *charge* means an algebraic operator, as in a differential or supersymmetry charge, not an electric charge or a selected particle quantum number. The Hodge-square mechanism has established precedents in supersymmetric quantum mechanics [89]. Its use here supplies a constrained source architecture, not a claim that the mechanism itself is new.

## 5.1 The shared center must be specified as a representation

On a qutrit basis  $e_j$ ,  $j \in \mathbb{Z}_3$ , put

$$Xe_j = e_{j+1}, \quad Ze_j = \omega^j e_j, \quad \omega^3 = 1, \quad \omega \neq 1.$$

Then  $XZ = \omega^{-1}ZX$  and  $XZX^{-1}Z^{-1} = \omega^{-1}I$ . The Fourier matrix with entries  $\omega^{jk}$  conjugates  $X$  to  $Z$  and  $Z$  to  $X^{-1}$ . On  $(\mathbb{C}^3)^{\otimes 3}$ , placing  $X, Z$  separately in each factor gives three commuting lanes with the same scalar commutator  $\omega^{-1}I_{27}$  [60].

The representation qualification matters. These are independent lane actions with their scalar centers identified, a central-product construction. They are not the diagonal tensor action of one qutrit group. For that diagonal action the two operators  $X^{\otimes 3}, Z^{\otimes 3}$  commute, since their commutator is  $\omega^{-3}I_{27} = I_{27}$ . Thus even the same tensor dimension does not specify the central character. The witness does not identify this carrier with CLN, the physical finite 27-dimensional carrier, or an exceptional Jordan algebra. Nor does an order-three scalar action alone construct the shared differential line with its continuum connection and holonomy.

## 5.2 Nilpotence links reversible transport and damping

Let  $d$  act on a finite-dimensional complex Hilbert space, with  $d^2 = 0$ . For a unit phase  $\zeta$ , define

$$B_\zeta = \zeta d + \bar{\zeta} d^\dagger, \quad \Delta = dd^\dagger + d^\dagger d.$$

The square expansion gives  $B_\zeta^\dagger = B_\zeta$ ,  $B_\zeta^2 = \Delta \geq 0$ , and  $\ker \Delta = \ker d \cap \ker d^\dagger$ . Let  $P_h$  project onto this harmonic space. If  $Z_h$  is skew-adjoint and commutes with  $d$  and  $d^\dagger$ , then for  $\gamma \geq 0$  the generator

$$\mathcal{L} = -iB_\zeta - \gamma\Delta + Z_h P_h$$

has mutually commuting summands. Consequently

$$e^{s\mathcal{L}} = e^{sZ_h P_h} e^{-isB_\zeta} e^{-\gamma s\Delta}, \quad \frac{d}{ds} \|\psi\|^2 = -2\gamma \langle \psi, \Delta \psi \rangle.$$

This is the single-charge factorization result [67]. It derives compatibility between the reversible operator and damping from  $d$ , while leaving  $d$ , its metric,  $\zeta$ ,  $\gamma$ , and  $Z_h$  as supplied data. The parameter  $s$  is not automatically physical time. Continuum extensions need domains and strong commutation, not just formal commutator identities.

For a small example take  $d = E_{12}$  on  $\mathbb{C}^3$ . Then  $\Delta = \text{diag}(1, 1, 0)$ . The first two modes decay for  $\gamma > 0$ ; on the third mode both  $B_\zeta$  and  $\Delta$  vanish. Persistent harmonic phase motion must therefore be supplied by  $Z_h P_h$ , not by the Hodge operator. This is compatible with the earlier strict-gradient circulation obstruction: the undamped motion belongs to the extra skew generator, not to gradient repair.

Changing  $\zeta$  changes the unsquared operator in a fixed frame, but that is not by itself an observable distinction. When  $d$  raises an integer grading  $N$  by one,

$$U_\theta = e^{i\theta N}, \quad B_{e^{i\theta}} = U_\theta B_1 U_\theta^\dagger.$$

If states and observables are transported as well, these are unitarily equivalent descriptions. A physical relative phase requires fixed comparison data. Squaring alone selects neither that data nor an arrow of time.

### 5.3 Curved squares can see phase without identifying it

For an arbitrary finite operator  $d$ , put  $\Delta_0 = dd^\dagger + d^\dagger d$ . The same expansion gives

$$B_\zeta^2 = \Delta_0 + \zeta^2 d^2 + \bar{\zeta}^2 (d^\dagger)^2.$$

The family is independent of *every* unit phase exactly when  $d^2 = 0$ . Its two nonconstant Fourier coefficients are  $d^2$  and  $(d^\dagger)^2$ ; both must vanish for constancy. But the sign pair  $\zeta, -\zeta$  always gives the same square [36].

There can be further ambiguities. For

$$d = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad d^2 = 2I, \quad \Delta_0 = 5I,$$

one obtains  $B_{e^{i\theta}}^2 = (5 + 4 \cos 2\theta)I$ . The squares at  $\theta = 0$  and  $\pi/2$  are  $9I$  and  $I$ , but  $\theta$  and  $-\theta$  give the same square and need not differ by a sign phase. Phase dependence is therefore not unique recovery even modulo sign.

For the graded Maurer–Cartan source the related identity is

$$d_a = d + [a, -], \quad d_a^2 = [F(a), -], \quad F(a) = da + a^2,$$

with graded brackets [13]. Here the operator curvature is the *action* of  $F$ , not necessarily the element  $F$  itself. Central curvature can be nonzero while its adjoint action vanishes. Ordinary cohomology is unavailable when  $d_a^2 \neq 0$ , but nonzero curvature does not forbid it in a representation where that curvature acts trivially. The curved-square correction is also not automatically the residual-Hessian correction of theorem 2.1; relating them requires their common source, pairing and admissible variations.

### 5.4 Jordan and Lie readouts lose different information

In a finite unital matrix star algebra with positive cyclic trace  $\tau$ , take  $F = F^\dagger$  and define

$$C_F(X) = FX + XF, \quad A_F(X) = FX - XF.$$

Together they reconstruct  $FX = (C_F X + A_F X)/2$  and  $XF = (C_F X - A_F X)/2$ . The Jordan map  $C_F$  is self-adjoint on the real Hermitian slice. The commutator is a derivation;  $-iA_F$ , not  $A_F$  itself, maps Hermitian observables to Hermitian observables and generates their unitary conjugation [32].

For a residual whose second derivative is  $XY + YX$ , the trace identity

$$\tau(F(XY + YX)) = \tau((FX + XF)Y)$$

identifies the ambient residual-Hessian correction with  $C_F$  on this real slice. For instance  $F = \text{diag}(2, 1)$  and  $X = E_{12} + E_{21}$  give  $C_F X = 3X$ ,  $A_F X = E_{12} - E_{21}$  and a Jordan quadratic pairing of six. This finite trace model does not automatically supply the graded signs, positive real slice or cyclic pairing of a physical q79 theory. A gauge row, a curved admissible stratum or additional physical residuals can also change its Hessian.

The information loss is exact in  $\mathcal{A} = \bigoplus_{\alpha=1}^r M_{n_\alpha}(\mathbb{C})$ . Write

$$F = F_0 + z, \quad z = \bigoplus_{\alpha} \lambda_{\alpha} I_{n_{\alpha}}, \quad \lambda_{\alpha} = \frac{\text{Tr } F_{\alpha}}{n_{\alpha}},$$

where each block of  $F_0$  is traceless. The split is orthogonal for a positive weighted block trace. Knowing  $[F, X]$  for *all*  $X \in \mathcal{A}$  determines  $F$  only modulo this center, whereas  $F = \frac{1}{2}C_F(1)$  recovers it from the complete Jordan map [29].

This last qualification is essential. A Hessian known only on an admissible tangent space may not include the unit or determine  $C_F(1)$ ; restricted commutator observations can likewise leave more ambiguity than the center. Complete ambient readouts cannot be assumed from a few projected rows.

An exact group action permuting isomorphic blocks leaves one invariant real central coordinate per block orbit. For  $M_2 \oplus M_2$ , the two center coordinates reduce to one under an imposed block exchange. That counts the remaining *central ambiguity*, not all source parameters. The algebra, symmetry, trace, admissible variations and traceless data remain inputs until selected. A one-dimensional center of  $M_{27}$  is not a theorem that the physical 27-carrier model has one free constant, or that the unresolved scalar rows are central curvature.

## 5.5 Admissible geometry comes before physical stability

For an odd self-adjoint finite charge  $Q$ , consider the specific homogeneous cost

$$F = Q^2, \quad E(Q) = \frac{1}{2} \|Q^2\|^2, \quad L_Q(a) = Qa + aQ.$$

With a fixed positive trace pairing,  $dE_Q(a) = \langle F, L_Q(a) \rangle$ . If the admissible tangent space  $V$  contains the radial direction  $Q$ , stationarity would require

$$0 = dE_Q(Q) = 2 \|F\|^2,$$

so nonzero curvature cannot be stationary in that radial direction [24]. This is a no-go for this homogeneous cost and variation space, not for every nonlinear defect. Fixing a differential's principal symbol, a normalization stratum or boundary data may remove the radial variation, but that restriction is source data and must be justified.

On a smooth admissible submanifold  $S$ , tangent restriction of the ambient Hessian alone is insufficient. With  $\Pi$  its normal second fundamental form,

$$\text{Hess}_S(E|_S)(a, b) = \text{Hess}_{\text{amb}} E(a, b) + \langle \text{grad}_{\text{amb}} E, \Pi(a, b) \rangle.$$

This is the same covariant chain rule as [theorem 2.1](#), now applied to the constrained source geometry. The extra term may remain nonzero at a critical point of  $E|_S$ , because that point need not be ambient-critical.

The finite circle witness takes  $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  and

$$Q_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad Q_1 = \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix}.$$

Both are self-adjoint and odd for grading  $\text{diag}(I_2, -I_2)$ ;  $Q_0^2 = Q_1^2 = I_4$  and  $Q_0 Q_1 + Q_1 Q_0 = 0$ . Thus  $Q_\theta = \cos \theta Q_0 + \sin \theta Q_1$  has constant square  $I_4$  and cost two under the ordinary trace. At zero, the ambient Hessian along  $Q_1$  is eight, while the normal-acceleration term is minus eight. The intrinsic circle Hessian is zero.

The circle therefore carries a protected first-order coordinate, not a damped mode proved by its positive ambient entry. This also does not select its traversal speed, identify it with the physical shared line, or turn its parameter into Lorentzian time. The useful synthesis is a source with an operator, metric, admissible variation geometry and shared representation, from which compatible readouts can follow. None of those source data can be reconstructed merely by renaming their projections.

## 6 Hodge, Green, and kernel shadows

### 6.1 The total-space Hodge package

Choose a smooth Hermitian metric on  $E^\bullet$ , a Hermitian metric on the compact boundaryless base  $J$ , and the associated density. Define

$$B_E = \overline{\mathbb{E}} + \overline{\mathbb{E}}^\dagger, \quad \Delta_E = B_E^2. \quad (6.1)$$

Its principal symbol is the Dolbeault symbol. Standard elliptic theory then gives a self-adjoint closure with compact resolvent [1].

**Theorem 6.1** (Hodge and kernel package). *Under the preceding compactness, ellipticity, metric, and domain hypotheses, there exist:*

1. *a finite-dimensional harmonic space  $\ker \Delta_E$  and orthogonal projector  $P_0$ ;*
2. *a smoothing heat semigroup  $e^{-s\Delta_E}$  for  $s > 0$ ;*
3. *a reduced Green operator*

$$G_E = (\Delta_E|_{\ker \Delta_E^\perp})^{-1}(I - P_0) = \int_0^\infty (e^{-s\Delta_E} - P_0) \, ds; \quad (6.2)$$

4. *finite-rank spectral projectors  $\chi_I(\Delta_E)$  for bounded spectral intervals  $I$ ; and*
5. *local Schwartz kernels for the smoothing functional calculus.*

*Every item is determined by the same selected  $\Delta_E$ .*

The word “shadow” is literal but not mystical. The spectral resolution is the wave-like normal-mode description. The local kernel

$$K_s(x, y) = \langle x | e^{-s\Delta_E} | y \rangle$$

is the localized propagation-of-repair description. They are two representations of one operator. Neither representation, by itself, makes the semigroup a physical quantum evolution.

### 6.2 The wave–particle operator obtains a source candidate

The conditional wave–particle paper begins with a supplied positive operator  $A$  and forms

$$B_{\text{adm}} = P\chi(A)e^{-\tau_{\text{adm}}A}\chi(A)P[34].$$

At the present structural tier, [theorems 4.2](#) and [6.1](#) provide the exact candidate

$$A = \Delta_1 = D\Phi_{\text{MC}}(0)^\dagger D\Phi_{\text{MC}}(0) \quad (6.3)$$

on the cohesive benchmark. This closes the formerly missing mathematical arrow from nonlinear repair to the positive filtered operator on that benchmark.

It does not close the physical common-source theorem of the wave–particle paper. That stronger theorem must show that the state, Hamiltonian or generator, effects, instrument, damping channel, and selected MTT source all belong to one physical construction. Equation (6.3) supplies the positive repair operator, not the rest of quantum mechanics.

### 6.3 A stratified warning

The existence of a regular total-space operator does not imply a uniform family of fiberwise inverses. The following exact example is the useful model.

**Proposition 6.2** (Regular complex, divergent fiberwise Green operator). *Let*

$$d_r = \begin{bmatrix} \text{diag}(1, 0, 0) & rI_3 \end{bmatrix} : \mathbb{C}^6 \longrightarrow \mathbb{C}^3$$

and let

$$B_r = \begin{bmatrix} 0 & d_r^\dagger \\ d_r & 0 \end{bmatrix}, \quad \Delta_r = B_r^2.$$

The entries of  $B_r$  and  $\Delta_r$  are polynomial in  $r$  and remain finite at  $r = 0$ . For  $r \neq 0$ ,

$$\text{spec}(\Delta_r) = \{0^{(3)}, (1+r^2)^{(2)}, (r^2)^{(4)}\}.$$

At  $r = 0$ ,

$$\text{spec}(\Delta_0) = \{0^{(7)}, 1^{(2)}\}.$$

Hence the smallest positive fiberwise eigenvalue is  $r^2$  and the reduced fiberwise Green norm grows as  $r^{-2}$ .

*Proof.* The three rows of  $d_r$  are orthogonal with squared norms  $1+r^2, r^2, r^2$ . These are the nonzero eigenvalues of both  $d_r d_r^\dagger$  and  $d_r^\dagger d_r$ , with each nonzero singular value appearing in both blocks of  $\Delta_r$ . The stated multiplicities and inverse norm follow directly.  $\square$

At  $r = 2$ , the spectrum is  $\{0^{(3)}, 4^{(4)}, 5^{(2)}\}$ ; at  $r = 0$ , the kernel jumps from dimension three to seven. The full complex is regular while the open-stratum contraction using  $1/r$  is not. This is why the cohesive object cannot be replaced globally by its rank-three open-stratum cohomology.

## 7 Descent, holonomy, and compatible projection

### 7.1 Nonlinear Čech descent

Let  $\{\mathcal{M}_i\}$  be local Riemannian configuration charts with local defect spaces  $\mathcal{E}_i$ . Suppose the overlap maps

$$U_{ij} : \mathcal{M}_i \rightarrow \mathcal{M}_j, \quad V_{ij} : \mathcal{E}_i \rightarrow \mathcal{E}_j$$

are respectively Riemannian isometries and unitary maps, and

$$\Phi_j \circ U_{ij} = V_{ij} \circ \Phi_i. \tag{7.1}$$

**Theorem 7.1** (Nonlinear repair descent). *The local costs  $\mathcal{C}_i = \frac{1}{2} \|\Phi_i\|^2$ , negative-gradient fields  $X_i = -\text{grad } \mathcal{C}_i$ , and their local flows  $R_i(s)$  obey*

$$\mathcal{C}_j \circ U_{ij} = \mathcal{C}_i, \tag{7.2}$$

$$TU_{ij}X_i = X_j \circ U_{ij}, \tag{7.3}$$

$$U_{ij} \circ R_i(s) = R_j(s) \circ U_{ij}. \tag{7.4}$$

Therefore the nonlinear Čech equalizer

$$\Gamma(\mathcal{M}) = \{(x_i) : x_j = U_{ij}x_i \text{ on every overlap}\}$$

is preserved. Compatible fixed points, basins, and existing asymptotic limits descend. A single-valued global basin label must lie in

$$\bigcap_{\gamma} \text{Fix}(\text{Hol}_{\gamma})$$

for the induced label local system.

*Proof.* Equation (7.1) and unitarity give equality of the local costs. Differentiating that equality and using that  $U_{ij}$  is an isometry intertwines the gradient fields. Uniqueness for the local initial-value problem then intertwines the flows. Passing to an existing limit commutes with the continuous overlap maps. Transport around a cycle gives the holonomy-fixed condition on a global label.  $\square$

This theorem does not select one basin. It says exactly when local repair is one global process. Unique actualization requires an additional theorem that one allowed global basin is selected, or an explicit probability law when several remain. Measurement need not be a privileged primitive here; it is one physical process capable of changing the local defect map and its basins.

## 7.2 Tangent and spectral projection

For a finite connected cover with unitary local-system transitions, the linear equalizer is concretely

$$\Gamma(\mathcal{H}) = \ker d_0, \quad (d_0\psi)_{ij} = \psi_j - U_{ij}\psi_i.$$

Transport along a spanning tree determines all components from one root vector; the remaining edges impose the cycle-holonomy fixed equations. Thus the global space is the intersection of those fixed spaces, not a tensor product over overlapping charts. Projectors descend only when  $U_{ij}P_i = P_jU_{ij}$ . These are the finite linear conclusions of the local Čech–Hilbert record cited in the evidence section.

Differentiate (7.1) at a compatible zero-defect fixed point. The Jacobians and Hessians intertwine, so the tangent heat semigroup preserves the linear Čech equalizer. The Hilbert-space description is thus the linearization of nonlinear descent, not its replacement.

**Theorem 7.2** (Compatible projection of the repair calculus). *Let  $H \geq 0$  be a self-adjoint repair Hessian on a Hilbert space and let  $P$  be an orthogonal projector reducing  $H$ . Put  $H_P = H|_{\text{Ran}P}$ . Then, on their declared domains,*

$$Pe^{-sH} = e^{-sH_P}P, \quad P\chi_I(H) = \chi_I(H_P)P, \quad PG = G_PP, \quad (7.5)$$

where the Green identity is taken on the corresponding harmonic complements. A unitary equivalence  $T$  of configuration spaces and a unitary equivalence  $U$  of residual spaces, satisfying  $\Phi'(Ta) = U\Phi(a)$ , transport the nonlinear cost jet and the tangent functional calculus at compatible zero-defect backgrounds.

For an isometric embedding rather than an onto equivalence, the cost jet is transported only as a pullback. Full tangent transport additionally requires, on the common operator domains,

$$J'T = UJ, \quad J'^{\dagger}U = TJ^{\dagger}.$$

These identities make the embedded tangent range reducing for the self-adjoint Hessian. Without the second identity only  $T^{\dagger}H'T = H$  is implied, not  $H'T = TH$ .

*Proof.* The first three identities are consequences of the spectral theorem for a reducing subspace. The residual identity gives the pullback equality  $\mathcal{C}'(Ta) = \mathcal{C}(a)$  and hence equality of all pulled-back cost derivatives. At a zero-defect background the two displayed derivative identities give  $H'T = J^\dagger J'T = J^\dagger UJ = TJ^\dagger J = TH$ . For onto unitary maps the adjoint identity follows from the chain identity. For an embedding it is an extra hypothesis. The common-domain self-adjoint intertwining then gives the claimed functional calculus.  $\square$

For example, let  $\Phi(x) = x$ ,  $\Phi'(x, y) = x + y$ ,  $Tx = (x, 0)$ , and  $U = 1$ . The residual and cost pull back exactly, but

$$H = 1, \quad H' = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad H'Tx = (x, x) \neq (x, 0) = THx.$$

The transverse response prevents reduction. This explains why a dimension match, an isometric compression or a compatible cost restricted to a subspace cannot replace a full operator intertwiner.

**Retained projectors are not the whole ambient projector.** There is a second, independent completeness requirement in the older continuum-to-finite cutset record [56]. Even exact reducing intertwining gives  $\Pi'T = T\Pi$ , not necessarily  $\Pi' = T\Pi T^\dagger$ . For a concrete counterexample, take

$$H = \text{diag}(0, 1), \quad H' = \text{diag}(0, 1, 0), \quad T(x, y) = (x, y, 0).$$

The intertwining residual is zero, both positive spectral gaps are one, and the range of  $T$  is reducing. Nevertheless,

$$\Pi' - T\Pi T^\dagger = \text{diag}(0, 0, 1), \quad \|\Pi' - T\Pi T^\dagger\| = 1.$$

Thus the packet's unrestricted whole-projector estimate  $\|\Pi' - T\Pi T^\dagger\| \leq 2\epsilon_H/g$ , based only on  $\epsilon_H = \|H'T - TH\|$  and a gap  $g$ , is false. It must not be used to promote the ambient kernel. In the exact reducing case the precise identity is

$$\Pi' = T\Pi T^\dagger + \Pi'_\perp,$$

where  $\Pi'_\perp$  projects onto the zero modes in  $(\text{Ran } T)^\perp$ . Equality of the whole projectors requires  $\Pi'_\perp = 0$ ; the retained intertwining identity needs no such assumption.

For approximate comparisons, the spectral object must also be fixed correctly. A small positive perturbation of a zero eigenvalue changes its literal kernel, although an isolated spectral cluster can persist. On a common Hilbert space let  $A, B$  be self-adjoint, let  $A - B$  extend to a bounded operator on their common operator domain, and let a common rectifiable resolvent contour  $\Gamma$  isolate the two clusters being compared. Writing  $M_A = \sup_{z \in \Gamma} \|(z - A)^{-1}\|$ , and similarly  $M_B$ , the resolvent identity and contour integral give

$$\|\Pi_A - \Pi_B\| \leq \frac{\text{length}(\Gamma)}{2\pi} M_A M_B \|A - B\|.$$

Indeed, subtract the two Riesz integrals and insert  $(z - A)^{-1} - (z - B)^{-1} = (z - A)^{-1}(A - B)(z - B)^{-1}$ . This is a cluster-comparison estimate, not a bound derived from an embedding residual alone. Identifying either cluster with the full physical harmonic sector requires a separate completeness certificate.

For a non-isometric Fourier–Mukai or BHT transform, derived equivalence alone is insufficient. It may preserve Ext groups, Yoneda products, and the formal Maurer–Cartan deformation problem after dg enhancement [19, 86, 6]. It does not automatically preserve the Hermitian metric, adjoint, Hodge spectrum, or numerical kernel. Those require a transported metric theorem or a quantified comparison defect.

The transported-metric result makes the latter task concrete. For a bounded invertible chain/product map  $T$ , with domains also carried by  $T$ , set

$$g_{\text{tr}} = (T^{-1})^\dagger g T^{-1}.$$

Then  $T$  is unitary for this transported metric. If a separately selected physical metric is  $g_{\text{phys}}(x, y) = g_{\text{tr}}(x, Ey)$ , its positive distortion  $E$  changes the degree- $k$  adjoint by

$$D_k^{\dagger_{\text{phys}}} = E_k^{-1} D_k^{\dagger_{\text{tr}}} E_{k+1}, \quad C_k = E_{k+1} D_k - D_k E_k.$$

Thus  $C_k = 0$  is the metric-chain compatibility test. The exact witness in the transported-metric record has transported Hodge matrix  $I_2$  but physical matrix  $\text{diag}(4, 1/9)$ . Polar normalization does not automatically repair the product: if  $T = US$ , its positive factor must itself respect the differential and product for  $U$  to be a dg-algebra map. These are explicit comparison tests, not a construction of the physical BHT operator or its HYM metric.

## 8 What survives a projection: compression, elimination and transfer

A retained space can inherit an operator without inheriting its entire algebra. Agreement of quadratic forms, products, Green operators and interaction vertices are different requirements. The following comparison integrates the finite compression and transfer results without identifying their carriers with a physical mass matrix. The general transfer machinery is standard homological algebra [21, 23]; the source-specific content is the exact finite realization and its constraints.

### 8.1 Leakage measures the failure of an algebra to descend

Let  $P$  be an orthogonal projector,  $Q = I - P$ , and write  $\Phi_P(S) = PSP|_{P\mathcal{H}}$ . For bounded operators,

$$\Phi_P(S)\Phi_P(T) - \Phi_P(ST) = -PSQTP.$$

Insert  $I = P + Q$  between  $S$  and  $T$  to prove the identity. The missing term is an excursion out of the retained space and back. Compression is multiplicative on a chosen algebra only when its excursions vanish, for example when the space reduces that algebra.

For commuting self-adjoint  $A, B$ , put  $L_A = QAP$  and  $L_B = QBP$ . Then

$$[\Phi_P(A), \Phi_P(B)] = L_B^\dagger L_A - L_A^\dagger L_B, \quad \|[\Phi_P(A), \Phi_P(B)]\| \leq 2 \|L_A\| \|L_B\|.$$

The exact three-dimensional witness takes  $Q = \frac{1}{3}\mathbf{1}\mathbf{1}^\top$ ,  $A = \text{diag}(1, 0, 0)$ ,  $B = \text{diag}(0, 1, 0)$ . Its compressed commutator, embedded in the ambient space, is

$$\frac{1}{9} \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

Commuting observables can therefore acquire a noncommutative compression. This does not prove that every projection does so, or derive canonical commutation relations, a Born state or Planck's constant [33].

The gauge-descent example supplies a concrete origin for this projector. On  $\mathbb{R}^3$ , set  $H_0 = 2P + 5Q$  and  $F(u) = H_0 u + (u^\top u)u$ , with repair equation  $\dot{u} = -F(u)$ . The inner product with  $u$  shows that  $F(u) = 0$  only at zero. The tangent operator is  $H_0$ ; its isolated eigenvalue-two space selects  $P$ . Signed permutations preserve the construction. Their conjugation action on the retained full matrix algebra has kernel  $\{\pm I\}$ , leaving a faithful quotient of order six. In general one must use the stabilizer of the chosen fixed point and an isolated spectral subspace. This is a descent mechanism, not selection of the Standard Model gauge group [61].

## 8.2 The internal operator cannot be omitted

For  $R = QRQ$ , write

$$E_R(S, T) = PS RTP, \quad E_R(S, T) - E_Q(S, T) = PS(R - Q)TP.$$

Raw compression uses  $R = Q$ . Feshbach elimination uses the inverse of  $Q(H - z)Q$  on  $Q\mathcal{H}$ , when it exists, and gives

$$P(H - z)P - PHQ[Q(H - z)Q]^{-1}QHP.$$

For  $P$  the first two coordinates of

$$H = \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & 2 & 0 & 0 \\ 1/2 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix},$$

the raw  $PHQHP$  is  $\text{diag}(1/4, 0)$ , whereas the inverse-weighted term at zero is  $\text{diag}(1/12, 0)$ . The effective operator is  $\text{diag}(11/12, 2)$ .

Homotopy transfer inserts a different object again:  $h = d^\dagger G$ , of degree minus one, rather than a degree-zero projector or resolvent. Its ternary operation has three inputs and two signed tree terms. An exterior differential algebra makes the distinction tangible: take  $da = db = 0$ ,  $dc = ab$ , an orthonormal monomial basis and harmonic complement spanned by  $ab, c$ . Here  $G = Q$ , but  $h(ab) = c$ ; the source sign convention gives  $m_3(a, a, b) = ac$ . Equality of  $G$  and  $Q$  has not removed the differential or changed a ternary map into a binary compression defect. This Heisenberg differential-algebra example is an auxiliary witness, not the global q79 topology [31].

## 8.3 Covariance completion changes the harmonic content

Start with  $A = M_3(\mathbb{Q}(\omega))$ ,  $\omega^2 + \omega + 1 = 0$ , a qutrit Weyl pair  $ZX = \omega XZ$ , and commuting automorphisms  $\alpha = \text{Ad } X$ ,  $\beta = \text{Ad } Z$ . The two-direction crossed exterior calculus has

$$K_W = A \oplus (A\theta_x \oplus A\theta_z) \oplus A\theta_x\theta_z, \quad \dim K_W = (9, 18, 9).$$

Its relations are  $\theta_x a = \alpha(a)\theta_x$ ,  $\theta_z a = \beta(a)\theta_z$ , with anticommuting square-zero forms, and  $da = (\alpha(a) - a)\theta_x + (\beta(a) - a)\theta_z$ . The finite metric gives Weyl-mode Laplacian eigenvalues 0, 3, 6, with multiplicities 1, 4, 4 before tensoring by forms. Harmonic dimensions are (1, 2, 1).

Their scalar-coefficient representatives form an exterior subalgebra, so transfer to these harmonics has  $m_n = 0$  for  $n \geq 3$ . A Hodge package alone does not manufacture nonzero interactions [65].

On harmonic degree one reflection acts as  $-I_2$ , and the quarter-turn as  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ . The associated determinant-twisted rank-six local-system construction has a rank-two Reynolds sector and four complementary strain directions. But the selected full-chain reflection and quarter-turn fail to preserve the old product. Cohomological compatibility is not full differential graded algebra (DGA) compatibility [59].

Closing the direction orbit under inversion and the quarter-turn gives  $+x, -x, +z, -z$ , hence a symmetric DGA  $A_{\text{sym}}$  with degree dimensions  $(9, 36, 54, 36, 9)$ , totaling 144. Squared norm  $1/2$  per signed edge retains the eigenvalues  $0, 3, 6$ . The finite order-36 group  $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes C_4$  now acts by DGA automorphisms. The old complex embeds isometrically as a reducing cochain summand, but not as a product subalgebra; its orthogonally compressed product is nonassociative off harmonics [74].

The completion has harmonic dimensions  $(1, 4, 6, 4, 1)$ , not  $(1, 2, 1)$ . The twelve extra classes exemplify the ambient modes that a retained intertwiner cannot exclude. These complexes are not quasi-isomorphic. Their dimensions are neither spacetime dimension, the 27-dimensional finite carrier, nor the rank-102 continuum response bundle.

#### 8.4 A first-jet quotient is not a deletion theorem

Let  $O$  span the odd differences  $o_i = \theta_{+i} - \theta_{-i}$ , and  $E$  the even sums  $e_i = \theta_{+i} + \theta_{-i}$ , for  $i = x, z$ . Formal connection transport along one direction gives

$$\frac{(e^{h\nabla_i} - 1)\theta_{+i} + (e^{-h\nabla_i} - 1)\theta_{-i}}{h} = \frac{\sinh(h\nabla_i)}{h}o_i + \frac{\cosh(h\nabla_i) - 1}{h}e_i.$$

The first jet is  $\nabla_i o_i$ ; the even channel starts with  $h\nabla_i^2 e_i/2$ . Reversal selects  $O$  as the universal first-symbol landing plane, with unique orthogonal projector  $(I - f^2)/2$  for the quarter-turn  $f$ . This formal identity does not construct a small-spacing family for the finite qutrit translations [66].

The exact harmonic quotient is

$$\mathcal{J} = \langle E \rangle \subset \Lambda(O \oplus E), \quad \Lambda(O \oplus E)/\mathcal{J} \simeq \Lambda(O).$$

The ideal has dimensions  $(0, 2, 5, 4, 1)$ , totaling twelve. This is a strict associative quotient of the harmonic algebra, not the nonassociative compression of the old full cochain complex.

Scaling  $o_i \mapsto o_i$ ,  $e_i \mapsto he_i$  defines a family of algebra maps. At nonzero scalar  $h$  it is invertible; at zero its image has dimension four and kernel  $\mathcal{J}$ . This statement about specialized images is not a flat family whose fiber dimension drops from sixteen to four. Specialization of the image module over  $\mathbb{Q}(\omega)[h]$  need not equal the image of the specialized map. The quotient and suppression order are exact; a flat Rees degeneration requires a separate construction. Higher-jet data are not thereby physically absent modes.

#### 8.5 The response retract keeps the missing classes

An actual strong deformation retract keeps these classes instead of discarding cohomology. Its target is

$$T = K_W \oplus \mathcal{J}, \quad \dim T = (9, 20, 14, 4, 1), \quad \dim T = 48.$$

For inclusion  $i$ , projection  $p$  and degree-minus-one homotopy  $h$ , the certified identities are

$$pi = I_T, \quad dh + hd = I - ip, \quad h^2 = ph = hi = 0.$$

The target differential  $m_1 = d_T$  is not zero. This is a response model, not a minimal model consisting only of cohomology. Transfer gives  $m_2 = p\mu(i \otimes i)$  and, in the source convention,

$$m_3(a, b, c) = p\mu(h\mu(ia, ib), ic) \\ - (-1)^{|a|} p\mu(ia, h\mu(ib, ic)).$$

The arity-three identity expresses the associator of  $m_2$  as the corresponding differential of  $m_3$ : strict associativity has a controlled chain-level replacement [72].

The frozen computation evaluates every target basis pair and triple and then all degree-admissible quadruples. Both  $m_3, m_4$  are nonzero; the arity-three and arity-four identities pass exactly. The quadruple audit checks 3,869,500 degree-admissible inputs, with 693,208 nonzero  $m_4$  values [73]. These archived computations are not rerun here. The nonzero inputs use response modes, so they do not contradict the vanishing of higher products on the harmonic subalgebra.

There is also an all-arity proof. In the fixed source basis take degree-one  $x, y, w$  and degree-zero  $z$ , labeled  $x = C_{0,0,1}$ ,  $y = C_{1,0,1}$ ,  $z = C_{1,0,0}$ ,  $w = C_{2,0,1}$ . The final index is an exterior-basis mask, not a third Weyl charge. A two-parity state recurrence proves

$$m_{2r+3}(x^{\otimes(2r+1)}, y, z) = \frac{2 + \omega}{4 \cdot 12^r} w, \\ m_{2r+4}(x^{\otimes(2r+2)}, y, z) = -\frac{\omega}{8 \cdot 12^r} w \quad (r \geq 0).$$

The hierarchy therefore does not truncate. A separate invariant-subspace argument gives  $m_n = 0$  for  $n \geq 3$  when at least  $n - 1$  inputs lie in  $\mathcal{J}$  [44]. These are not exhaustive tables: the cited snapshot does not compute the complete  $m_5$  table or enumerate every arity-five Stasheff residual. The transfer theorem supplies formal identities; exhaustive table verification is a distinct implementation check.

## 8.6 One source map can transport the whole hierarchy

Let  $\varphi : A \rightarrow A'$  be a degree-zero DGA morphism and  $\psi : T \rightarrow T'$  a retained chain map between normalized contractions. If

$$\varphi i = i' \psi, \quad p' \varphi = \psi p, \quad \varphi h = h' \varphi,$$

then  $\psi m_n = m'_n \psi^{\otimes n}$  for every  $n \geq 1$ . Move  $\varphi$  through each decorated transfer tree: its product identity handles vertices, the homotopy square handles internal edges, and the other squares handle leaves and root. Degrees, hence signs, are unchanged. This proves all-arity naturality rather than extrapolating from low-order tests [25].

The finite translation and Fourier generators preserve the complete 144-to-48 contraction. Their order-36 covariance therefore acts on every transferred operation. At the continuum level a same-source product-preserving map, reducing domains, metric and physical symmetry lift must still be supplied. Approximate squares require defect bounds; they do not give exact naturality.

The finite program thus has an exact response hierarchy and a finite set of sufficient source identities for transporting it. It has not selected the continuum HYM source or its Lorentzian/cyclic action. These  $m_n$  are not measured Yukawa or gauge couplings: they specify what must be preserved before that physical interpretation is justified.

## 9 Primitive repair as a quotient, not a chosen plane

The preceding projection results ask what survives after a complex is compressed. A complementary question occurs in the component-20 algebraic calculation: what is the correct target of the repair map itself? Eight source records, UST.G3GMS–G3GMZ, successively replace matching ranks, ordering conventions and embedded complements by a typed boundary map and an intrinsic quotient. Their proofs and archived arrays remain owned by the cited source records. Here we explain how their conclusions fit the repair language, rather than introducing eight duplicate theorems.

These are finite algebraic results. The campaign consists of eight primes with six selected roots each, hence 48 fibers. The complete ordering calculations in G3GMT and G3GMU are at the anchor  $k = \mathbb{F}_{21817}$ ,  $\gamma = 5$ . They are not automatically 48-fiber ordering theorems. The degree-15 polynomial carrier has dimension  $\binom{17}{2} = 136$ ; it is neither the finite 27-state carrier nor a physical mass matrix. *Repair* below first means an algebraic retraction, not a positive heat generator or a new spacetime dynamics.

### 9.1 The PBW correction identifies maps, not just ranks

The Poincaré–Birkhoff–Witt (PBW) calculation measures a defect in an ordered presentation of the selected response complex. Two earlier constructions had rank-six corrections in different spaces. Their equal ranks alone gave no reason to identify them. The G3GMS result supplies the missing typed identification [45]. With an injective response boundary  $N$ , invertible adjugate normalization  $R_H$ , and anomaly factor  $R_{Q,H}$ , write

$$N_H = N R_H, \quad A_{\text{PBW}} = R_H^{-1} R_{Q,H}, \quad N_H A_{\text{PBW}} = N R_{Q,H}. \quad (\text{P1})$$

The response splitting provides maps  $J, W, L_H$  with

$$[J \ N_H]^{-1} = \begin{bmatrix} W \\ L_H \end{bmatrix}.$$

Thus the lifted coefficient action is

$$\Lambda_H(A) = JAW + N_H A L_H = [J \ N_H] \text{diag}(A, A) [J \ N_H]^{-1}. \quad (\text{P2})$$

This explains why a rank-six coefficient action lifts to rank twelve, and why its algebraic relations survive. The archived response differential and homotopy commute with this lift.

At the anchor, the 36 row/column block orders yield distinct raw presentations but one response-boundary image. Explicit invertible six-by-six matrices relate their boundary bases. This does not say that the raw matrices or their transfer determinants coincide. At all 48 fibers the normalized anomaly and response lift have ranks six and twelve, respectively, with the recorded chart covariance. The common boundary image is the invariant at this stage, not a preferred ordering.

### 9.2 The ordering orbit has a canonical finite decomposition

G3GMT next constructs the left coefficient presentation [30]. Let  $F_o$  be the structural tail evaluator for an ordering  $o$ , and  $C_{qr}$  the concatenated  $q$ -principal,  $q$ -commutator and  $r$ -commutator blocks. For every anchor ordering,

$$\text{rank } F_o = 120, \quad \text{rank } C_{qr} = 18, \quad \text{rank}[F_o \ C_{qr}] = 136.$$

Consequently the presentation has a two-dimensional relation space:

$$0 \longrightarrow k^2 \longrightarrow k^{120} \oplus k^{18} \longrightarrow \text{Sym}^{15}(k^3) \longrightarrow 0. \quad (\text{P3})$$

The two-plane is a genuine overlap, not an extra physical freedom.

The source minimal polynomial of  $A_{\text{PBW}}$  is  $xf(x)$ , where  $f$  has degree six and  $f(0) \neq 0$ . Coprimality gives the Fitting split and its polynomial projector:

$$k^{18} = \ker(A_{\text{PBW}})_{12} \oplus \text{im}(A_{\text{PBW}})_6, \quad P = I - \frac{f(A_{\text{PBW}})}{f(0)}, \quad Q = I - P. \quad (\text{P4})$$

Here  $P$  is algebraic, not a Hilbert-space orthogonal projector.

G3GMU uses the supplied factorial-apolar symmetric pairing on the three quadratic blocks to connect these factors [26]. Writing an overlap basis as  $X_o$ , put  $K_o = QX_o$ ,  $Y_o = PX_o$ . The nondegenerate restricted Gram forms define, for example,

$$T_o = Y_o(K_o^t G K_o)^{-1} K_o^t G Q.$$

The reverse arrow is defined by exchanging the two factors. Averaging the 36 arrows, then normalizing the return on  $\text{im } P$ , gives

$$T^2 = S^2 = 0, \quad TS = P, \quad ST = R, \quad I = P + R + (Q - R). \quad (\text{P5})$$

All three summands have rank six;  $T + S$  is a partial involution,  $(T + S)^2 = P + R$ , rather than a square-zero differential. A square-zero mapping cone instead uses the lifted arrow and the correct opposite differential on its second diagonal block.

This distinction matters in the cohesive language: an off-diagonal extension can be exact without being the final coefficient compiler. The source-first tests reject seven elementary averaging/cross-term outputs in G3GMT, six direct-arrow outputs and nine central-character outputs in G3GMU. They reject those finite candidates, not all repair maps. The later quotient construction below is not a claim that a rejected right multiplication suddenly works.

### 9.3 From chart retraction to a tensor-natural extension

For the selected graded ideal  $I \subset k[x, y, z]$ , let  $I_{15}$  be its degree-15 part and  $K = I_{15}^\perp$  its inverse system, using the dual divided-power coordinates. The source ranks are

$$\dim I_{15} = 100, \quad \dim K = 36, \quad K = H \oplus C, \quad \dim H = \dim C = 18.$$

The inherited derivative space  $H$  is intrinsic to the supplied ideal and derivative filtration; the displayed complement  $C$  is initially a normalized chart. G3GMV constructs an idempotent ambient retraction onto that complement [64]. It reconstructs the 48 quotient bases before opening the comparison archive.

G3GMW improves the extension away from  $K$  [40]. With  $B = [H \ C]$ , use the supplied factorial-apolar form  $G_{15}$ , whose displayed monomial diagonal is  $a!b!c!$  for  $a + b + c = 15$ , and define

$$G_B = B^t G_{15} B, \quad F_B = [0 \ I_{18}] G_B^{-1} B^t G_{15}, \quad R_{\text{nat}} = C F_B. \quad (\text{P6})$$

Invertibility of  $G_B$  is a separate verified condition at every archived fiber; nondegeneracy of the ambient form alone would not imply it. All campaign primes exceed 15. The form is nondegenerate and symmetric over those finite fields, not a positive physical metric.

The identities  $F_B H = 0$ ,  $F_B C = I_{18}$  imply  $R_{\text{nat}}^2 = R_{\text{nat}}$  and rank eighteen. It is the specified projection on  $K$ , extended by zero on  $K^\perp_{G_{15}}$ . It need not be self-adjoint:  $H$  and  $C$

are not assumed orthogonal. Changing bases within  $H$  and  $C$  leaves the operator unchanged. An ambient change  $U$  requires transporting the pairing as a tensor:

$$B' = UB, \quad G'_{15} = U^{-t}G_{15}U^{-1} \implies R'_{\text{nat}} = UR_{\text{nat}}U^{-1}. \quad (\text{P7})$$

This is covariance of the full supplied datum, not invariance under arbitrary  $U$  while keeping a fixed numerical diagonal.

The actual anchor source return  $M_{\text{act}}$  satisfies  $\text{rank}(F_B M_{\text{act}}) = 18$ . Therefore

$$R_{\text{nat}} M_{\text{act}} (F_B M_{\text{act}})^{-1} = C.$$

Normalization recovers the source-derived quotient exactly. The 48 reconstructed quotient bases are not 48 independently executed actual source returns. This is an important distinction in interpreting the frozen comparison, even though both calculations use exact arithmetic.

## 9.4 The complement disappears in the quotient

The stronger target is  $\mathcal{Q} = K/H$ , not an embedded  $C$ . G3GMX proves the adapted change-of-frame law [27]. If

$$B' = UB \begin{bmatrix} A & E \\ 0 & D \end{bmatrix}, \quad G'_{15} = U^{-t}G_{15}U^{-1},$$

then

$$F_{B'} = D^{-1}F_B U^{-1}. \quad (\text{P8})$$

The mixing block  $E$  disappears because the bottom block row of the inverse adapted frame is  $[0 \ D^{-1}]$ . Thus no preferred global complement is needed. Once  $K, H$  are subbundles and the pairing and overlaps are compatible, these formulas give the quotient-map descent. They do not by themselves construct those subbundles in characteristic zero.

A small example clarifies what is and is not independent of splitting. Take  $K = k^2$ ,  $H = ke_1$ ,  $C = ke_2$ , and the identity pairing. The quotient coordinate is  $F = (0, 1)$ . Replacing  $e_2$  by  $e_2 + ae_1$  leaves  $F$  unchanged, but changes its ambient lift to

$$R_a = \begin{pmatrix} 0 & a \\ 0 & 1 \end{pmatrix}.$$

Every  $R_a$  is idempotent. The common output is the class modulo  $H$ , not a unique matrix of representatives. A physical orthogonal projector would require additional compatible structure.

The same source record also binds a concrete input carrier: for  $q = xz - y^2$ , the restriction

$$q^4 \text{Sym}^7(k^3) \longrightarrow K/H$$

is surjective at all 48 archived fibers. Its source has dimension 36 and target dimension 18, with an emitted kernel relation matrix. This is stronger than guessing eighteen ambient vectors and weaker than a global physical-source selection theorem.

## 9.5 The quotient is dual to the socle

G3GMZ identifies why this quotient occurs [28]. Put  $\mathcal{A} = k[x, y, z]/I$ , and  $K_d = \mathcal{A}_d^*$  in divided-power coordinates. The socle is the part annihilated by multiplication by all three variables. At degree 15,

$$\text{Soc}(\mathcal{A})_{15} = \ker[\mathcal{A}_{15} \xrightarrow{(x,y,z)} \mathcal{A}_{16}^{\oplus 3}].$$

Multiplication is dual to divided-power differentiation. Hence

$$\text{Soc}(\mathcal{A})_{15}^* \simeq K_{15}/(\partial_x K_{16} + \partial_y K_{16} + \partial_z K_{16}) = K/H. \quad (\text{P9})$$

Here  $\dim K_{16} = 11$ , while the combined derivative image has rank 18. Its three summands need not be independent. This dual pairing is the multiplication/derivative pairing; it should not silently be replaced by the auxiliary ambient Gram form in (P6). Divided-power conventions are described in the standard inverse-system documentation [20].

For intuition, in  $k[x, y, z]/(x, y, z)^2$  all degree-one classes are socle classes. There is no degree-two inverse system to differentiate, so the primitive degree-one quotient is the entire dual three-space. The selected component-20 quotient obeys the same kernel/cokernel logic, but with a nonzero inherited derivative space. Its rank eighteen is therefore explained by an algebraic duality, not by a particle count.

## 9.6 The terminal resolution supplies the actual comparison map

For an Artinian quotient of a three-variable polynomial ring, top Koszul homology gives  $\beta_{3,j} = \dim \text{Soc}(\mathcal{A})_{j-3}$ . The source terminal row is

$$R(-18)^{18} \oplus R(-19) \oplus R(-22), \quad R = k[x, y, z].$$

Thus the degree-18 terminal block matches the degree-15 socle in rank. G3GMZ supplies more than this numerical match: it emits a chain comparison at all 48 fibers [75].

For the selected free resolution  $F_\bullet$  and the Koszul complex  $K_\bullet(x, y, z)$ , use  $D = d_F + (-1)^p d_K$  on  $F_p \otimes K_q$ . A terminal generator  $u_3$  is completed to a total cycle through

$$d_K u_2 = -d_3 u_3, \quad d_K u_1 = d_2 u_2, \quad d_K u_0 = -d_1 u_1. \quad (\text{P10})$$

The final coefficient modulo  $I_{15}$  lies in the socle. The emitted eighteen classes are independent and pair nondegenerately with  $K/H$ . Forward and reverse choices of Koszul preimages give the same quotient classes. Their polynomial representatives need not agree. This is the appropriate comparison-up-to-homotopy principle, not a claim of unique contracting matrices [84].

Together, these records remove an arbitrary complement and a separate terminal-to-socle identification from the finite construction. They do not select a positive HYM Hessian, establish an analytic semigroup or derive a physical value. The general formulas apply beyond the finite campaign when their source complex, ranks and pairings are supplied. These eight certificates alone do not supply a symbolic common complex over  $\mathbb{Q}(\gamma)$  or the other 47 actual source returns. Historical open sentences in G3GMS–G3GMZ are read through the later refinements above, not reimposed as current research blockers. Broader promotion belongs to the current source ledger; this section records the precise contribution of these eight archived results.

## 10 Nested repair and curvature-preserving transfer

There are two different places where a differential can act: inside a configuration, and on the space of configurations. Keeping them distinct allows a curved internal charge to coexist with an ordinary outer complex. The six source records discussed here join that observation to the Heisenberg example in [section 8.2](#), then pass from central curvature to a positive noncentral coefficient algebra. They are finite or local structural results, not a new physical endpoint.

### 10.1 Inner curvature and an outer nilpotent differential

Let  $M$  be a smooth Riemannian configuration manifold with a smooth internal charge  $Q_x$ , defect  $\Phi(x) = Q_x^2$ , and a declared positive fiber metric. Put  $V = \frac{1}{2} \|\Phi\|^2$ , allowing an explicitly normalized trace in the finite examples below. On forms over  $M$ , define

$$d_{\beta,V} = e^{-\beta V} d_M e^{\beta V} = d_M + \beta dV \wedge, \quad \beta \geq 0.$$

This is Witten deformation [\[89\]](#). Conjugation gives  $d_{\beta,V}^2 = 0$  for every smooth  $V$ , whether or not the internal defect vanishes [\[47\]](#). Thus  $[Q_x, -]^2 = [\Phi(x), -]$  can be nonzero while the outer differential is nilpotent. These operators act on different spaces; neither identity repairs the other by changing its meaning.

For a closed compact  $M$ , the usual elliptic realization gives the positive operator

$$H_{\beta,V} = d_{\beta,V} d_{\beta,V}^\dagger + d_{\beta,V}^\dagger d_{\beta,V}$$

and its contractive heat semigroup. Boundary or noncompact settings need compatible adjoints and self-adjoint domains; the polynomial computations below are formal differential-operator identities, not domain certificates. On the configuration line let  $A_V = \partial_x + \beta V'$ . Then

$$\begin{aligned} H_0 &= A_V^\dagger A_V = -\partial_x^2 + \beta^2 (V')^2 - \beta V'', \\ H_1 &= A_V A_V^\dagger = -\partial_x^2 + \beta^2 (V')^2 + \beta V''. \end{aligned}$$

At a critical point,  $V''$  contains the full residual-Hessian term from [theorem 2.1](#), not just the Gram term. For  $\Phi(x) = 1 + x^2$ , one has  $V = \frac{1}{2}(1 + x^2)^2$  and, at zero,

$$\Phi = 1, \quad \Phi' = 0, \quad V' = 0, \quad |\Phi'|^2 = 0, \quad \Phi\Phi'' = 2, \quad V'' = 2.$$

The opposite potential terms  $-2\beta$  and  $2\beta$  are not opposite spectral-positivity claims: positivity belongs to the factorized differential operators, including their kinetic terms.

This construction retains the defect's full second jet, but it does not make the Witten operator equal to the internal curvature or to the finite-dimensional Hessian matrix. On a closed compact  $M$ , multiplication by  $e^{\beta V}$  identifies its cohomology with de Rham cohomology. Nilpotence alone therefore does not select zero defect or a unique physical configuration. The supplied  $\beta$  is a localization parameter, not Lorentzian time or a derived thermodynamic temperature.

### 10.2 What the eliminated pair leaves behind

For the auxiliary algebra  $\Lambda(a, b, c)$  with  $dc = ab$ , the six harmonic representatives are

$$H = \text{span}(1, a, b, ac, bc, abc).$$

Let  $i$  include them, let  $p$  remove the pair  $c, ab$ , and set  $h(ab) = c$ , with  $h = 0$  otherwise. Then

$$pi = I_H, \quad dh + hd = I - ip, \quad h^2 = ph = hi = 0.$$

Under the transfer convention already used in [section 8.5](#), the two nonzero examples are

$$m_3(a, a, b) = ac, \quad m_3(a, b, b) = -bc \quad [43].$$

They remember the removed relation  $dc = ab$ ; a cohomology-dimension count does not encode them. Minimal-model and strict-unital constructions are standard homological algebra [18, 8].

With weights  $w(a) = w(b) = 1$ ,  $w(c) = 2$ , every source operation and contraction map preserves total integer weight, hence also its reduction modulo three. This is a representation-level phase statement, not a construction of the continuum shared line or an identification of Heisenberg/Nil with the global q79 manifold.

The top-form functional  $\tau_A(u) = [abc]u$  is graded cyclic but is not a positive Hilbert norm. It also carries weight four: under the stated circle action,  $\tau_A(g_\theta u) = e^{4i\theta} \tau_A(u)$ . Thus its pairing is covariant with that character, not automatically an invariant scalar pairing. An invariant scalar interpretation needs the inverse-character line or equivalent compensating data. Preserving a cyclic contraction and preserving weights are compatible assertions; neither supplies that additional scalar trivialization.

### 10.3 Which data a nested contraction must preserve

A quasi-isomorphism is insufficient for this purpose. The restricted transfer record requires the same admissible base, typing and boundary data; maps  $p, i, h$  satisfying the contraction and side identities; first-order and adjoint intertwiners; compatible cyclic and Hilbert pairings; a unit and equal normalized repair potentials; and shared-line equivariance [62]. The maps must be parallel over the configuration base, with a common  $\beta$ , to intertwine the outer Witten operators. Transfer trees also need a terminating filtration or a proved convergence domain.

For two algebraically compatible contractions the composite is

$$p = p_2 p_1, \quad i = i_1 i_2, \quad h = h_1 + i_1 h_2 p_1.$$

These formulas preserve the normalized contraction identities. Analytic admissibility of the composite still needs a common transfer-control class; convergence on unrelated domains is not automatically enough. Likewise, local arrows alone do not prove effective overlap descent.

Here is the same-source central example. Let  $\Gamma_A$  be form parity, take  $S = \mathbb{R}^{2|2}$ , and define

$$R(x) = \begin{pmatrix} 1 & x \\ -x & 1 \end{pmatrix}, \quad C(x) = \begin{pmatrix} 0 & R(x) \\ R(x)^\top & 0 \end{pmatrix}, \quad Q_A = d \otimes I_4 + \Gamma_A \otimes C(x).$$

Since  $C(x)^2 = (1 + x^2)I_4$  and  $d\Gamma_A = -\Gamma_A d$ ,

$$Q_A^2 = (1 + x^2)I_{32}, \quad Q_H = \Gamma_H \otimes C(x), \quad Q_H^2 = (1 + x^2)I_{24}.$$

Tensor the preceding contraction with  $I_4$ . It intertwines  $Q$ , its declared adjoint and curvature. Rank-normalized trace gives  $V = \frac{1}{2}(1 + x^2)^2$  on both fibers; raw traces would differ with rank. Because  $p$  is independent of  $x$ , it also intertwines the formal outer Witten blocks. The same  $p, i, h$  produce the nonzero  $m_3$ .

Strict intertwining and transfer remain different claims: the source DGA has no ternary operation to project, while its minimal model has one. Equality of normalized costs is verified for this source, not inferred from any arbitrary rank-reducing map.

## 10.4 Two ways to retain noncentral curvature

The infinitesimal route works over  $D = \mathbb{Q}[\epsilon]/(\epsilon^2)$ , with matrix indices starting at zero and parities  $(0, 1, 0)$ . Put

$$q = E_{10} + \epsilon E_{21}, \quad F = q^2 = \epsilon E_{20}, \quad [F, E_{00}] = F \neq 0.$$

For homogeneous  $X$ , the graded commutator  $d_q X = qX - (-1)^{|X|} Xq$  satisfies  $d_q^2 X = [F, X]$  and  $d_q F = 0$ . Every two-curvature product  $FXF$  vanishes over  $D$ . For the Heisenberg contraction and  $\delta = \Gamma_A \otimes q$ ,

$$(h\delta)^2 = (\delta h)^2 = 0, \quad (I - h\delta)^{-1} = I + h\delta.$$

This is exact algebraic termination, not a small numerical error [48]. The 24-to-18 module ranks are over  $D$ , hence double over  $\mathbb{Q}$ . The coefficient-of- $\epsilon$  supertrace is nondegenerate and graded cyclic, not positive. No nonzero nilpotent self-adjoint operator can represent this  $F$  on a positive finite Hilbert space.

The positive route instead takes  $S = \mathbb{R}^{1|2}$  and

$$q(x) = \begin{pmatrix} 0 & 1 & x \\ 1 & 0 & 0 \\ x & 0 & 0 \end{pmatrix}, \quad F(x) = q(x)^2 = \begin{pmatrix} 1+x^2 & 0 & 0 \\ 0 & 1 & x \\ 0 & x & x^2 \end{pmatrix}.$$

Here  $q = q^\top$ ,  $F = q^\top q \geq 0$ , and  $[F(0), E_{12}] = E_{12}$ . The same form contraction gives  $Q_A = d \otimes I_3 + \Gamma_A \otimes q$  on 24 module directions and  $Q_H = \Gamma_H \otimes q$  on 18. It retains  $Q_A^2 = I_A \otimes F$ , adjoint intertwining and the normalized cost

$$V(x) = \frac{1}{3}(1+x^2)^2, \quad V''(0) = \frac{1}{3}\|F'(0)\|_{\text{HS}}^2 + \frac{1}{3}\langle F(0), F''(0) \rangle_{\text{HS}} = \frac{2}{3} + \frac{2}{3} = \frac{4}{3} \quad [51].$$

The omitted half would be a genuine error in a Gram-only Hessian. The positive norm uses ordinary trace; graded cyclicity uses supertrace. They are separate pairings, not one automatically positive cyclic form.

A wording correction is necessary here. The self-adjoint coefficient matrix is  $q$ , not the full module superconnection:

$$Q_A - Q_A^\dagger = (d - d^\dagger) \otimes I_3 \neq 0.$$

Indeed  $d(c) = ab$  while  $d^\dagger(c) = 0$  in the monomial metric. The positive square  $Q_A^2 = I_A \otimes F$  and all displayed intertwiners remain exact, but  $Q_A^2$  must not be renamed  $Q_A^\dagger Q_A$ . Neither this module dimension nor the central example's outer-form dimension identifies the different CBF response carrier, the physical 27-carrier, or the rank-102 bundle.

## 10.5 Curvature as retained coefficients

The positive curvature is not nilpotent. Nevertheless the strict-unital extension record closes the local curvature-insertion problem for this particular tensor construction [70]. Keep  $B = \text{End}(\mathbb{R}^{1|2})[x]$ , with  $d_B = [q, -]$  graded and  $d_B^2 = [F, -]$ , as the coefficient algebra. Transfer only the Heisenberg form factor.

For this six-class minimal model,  $m_n = 0$  for  $n \geq 4$ . A nonzero internal homotopy introduces  $c$ ; a later homotopy cannot remove it to recover  $ab$ , and two branches carrying  $c$  multiply to zero. Hence every tree with at least two homotopies vanishes. This is specific to the displayed exterior algebra and contraction, not a universal termination theorem.

Use total  $\mathbb{Z}_2$  grading for the supermatrix extension; the integer form grading alone does not give an integer grading of the finite matrices. On  $K = H \otimes B$ , the source convention is

$$\begin{aligned} M_0 &= -1_H \otimes F, \\ M_1(u \otimes b) &= (-1)^{|u|} u \otimes d_B b, \\ M_n(u_1 \otimes b_1, \dots, u_n \otimes b_n) &= (-1)^{\chi} m_n(u_1, \dots, u_n) \otimes b_1 \cdots b_n \quad (n = 2, 3), \\ \chi &= \sum_{i < j} |b_i| |u_j|, \quad M_n = 0 \quad (n \geq 4). \end{aligned}$$

The ordered coefficient product is essential;  $B$  is not commutative. At fixed  $x$ ,  $K$  has dimension  $6 \cdot 9 = 54$ , an operator algebra carrier rather than the 18-dimensional module above.

The curved arity-one identity reads

$$M_1^2(X) + M_2(M_0, X) - M_2(X, M_0) = 0.$$

Thus  $M_0 = -F$  is a sign convention, not a claim that positive curvature became a negative repair norm. In every higher curvature insertion the  $H$  factor is the strict unit, so  $M_n(\dots, M_0, \dots) = 0$  for  $n \geq 3$ . The remaining identities reduce to Bianchi, the graded derivation rule and the Heisenberg  $m_2/m_3$  identities through arity five. No term remains at higher arity. The frozen record checks the 20,736 and 248,832 form/parity cases at arities four and five; those tables are not rerun here.

The scalar-curvature notation  $m_0 = \lambda 1$  in the earlier finite example must likewise be interpreted in parity grading, or with a formal parameter of degree two if integer cohomological degrees are required. A real degree-zero scalar is not an integer-degree-two curvature merely because it is called even.

This closes the local algebraic conjunction of noncentral curvature, positive cost, outer nilpotent repair and transferred interactions. It does not require an infinite curvature-insertion sum in this model. There is no contradiction with the nontruncating CBF hierarchy of [section 8.5](#): that target is a different, nonminimal 48-dimensional response retract. A selected physical factorization, global descent, domains, shared differential line, signed action and normalization still require their own evidence. The examples explain what those data could organize; they do not select them.

## 11 Global descent and the actual rational topology

The local tensor examples answer how a coherent algebra can retain curvature under transfer. Two further records separate the global questions: how projective charts glue that algebra, and which rational form factor the q79 topology actually supplies. Neither question is answered by identifying every circle construction with the auxiliary Heisenberg example.

### 11.1 What descends through projective overlaps

Keep the projective atlas of [theorem 4.1](#). On a good cover, let  $G_{ij}$  be even unitary transitions satisfying  $G_{ij}G_{jk}G_{ki} = \alpha_{ijk}I$ , and suppose odd coefficient charges obey  $q_i = G_{ij}q_jG_{ij}^{-1}$ . Set

$$B_i = \text{End}(E_i), \quad F_i = q_i^2, \quad d_i = [q_i, -].$$

Conjugation is an honest algebra cocycle even though the module cocycle is projective. Squaring and taking graded commutators give

$$F_i = \text{Ad}(G_{ij})F_j, \quad d_i \text{Ad}(G_{ij}) = \text{Ad}(G_{ij})d_j.$$

Thus the earlier endomorphism untwisting extends to a curved differential, not just to the zero-curvature case [76]. The coefficient charge need not commute with the transitions; it need only transform correctly.

Suppose the form-factor contractions have honest overlaps  $T_{ij}$  preserving the unit and intertwining their transferred  $m_n$ . Then

$$R_{ij} = T_{ij} \hat{\otimes} \text{Ad}(G_{ij})$$

has an honest triple law. For the coefficient operations of [section 10.5](#),

$$R_{ij} M_{n,j} = M_{n,i} R_{ij}^{\otimes n}, \quad n \geq 0.$$

For  $n = 0$  this means transport of curvature. For  $n \geq 2$  it follows from transport of  $m_n$  and the ordered coefficient product; even transitions preserve the Koszul signs. Hence the local curved identities glue without introducing another curvature-insertion series. This imports the overlap extension; it does not duplicate the untwisting proof.

An ordinary endomorphism sheaf is not necessarily a globally split matrix algebra. In the locally free case it can be an Azumaya algebra with nontrivial Brauer class [9]. The scalar twist cancels from conjugation, not from the original module. For the same reason, central shared-line holonomy cannot be reconstructed from this adjoint action alone. The Hermitian differential line remains separate source data.

## 11.2 A noncommuting witness and its metric boundary

Let  $X, Z$  be the qutrit shift and clock, with  $Xe_j = e_{j+1}$ ,  $Ze_j = \omega^j e_j$ , and  $\omega^3 = 1$ . On  $\mathbb{C}^{1|2} \otimes \mathbb{C}^3$ , take

$$G_{01} = I_3 \otimes X, \quad G_{12} = I_3 \otimes Z, \quad G_{20} = \omega(G_{01}G_{12})^{-1},$$

so their product is  $\omega I_9$ , while the adjoint triple is the identity. A nonidentity triple in this chosen atlas does not by itself certify that a global gerbe class is nonzero. The source charge is

$$q_0 = (E_{01} + E_{10}) \otimes I_3 + E_{02} \otimes X + E_{20} \otimes X^\dagger.$$

It is odd and self-adjoint, and a clock overlap moves it nontrivially. The transformed charges glue, their  $F_i = q_i^\dagger q_i = q_i^2$  are positive and noncentral, and every chart has

$$V_i = \frac{1}{18} \text{Tr}(F_i^\dagger F_i) = \frac{4}{3}.$$

Here self-adjointness concerns the displayed coefficient  $q_i$ , not the full form-module operator corrected in [section 10.4](#). The witness tests actual conjugation, not a commuting-factor substitute.

There is also a useful limit to this example. Its Heisenberg overlap acts by  $\omega^w$ , where the top form  $abc$  has weight four. Consequently its top-form functional satisfies

$$\tau_A(TaTb) = \omega \tau_A(ab),$$

not scalar invariance. The finite check certifies algebraic equivariance and the triple law, not all scalar cyclic-pairing hypotheses of the general descent theorem. A line-valued pairing, or the inverse-character line, supplies the needed compensation. This does not invalidate the algebraic descent or the positive coefficient trace.

Likewise, coordinate-dependent transitions require the connection law, not only matrix conjugation. In a local trivialization it has the form

$$\mathcal{A}_i = G_{ij}\mathcal{A}_jG_{ij}^{-1} - (dG_{ij})G_{ij}^{-1},$$

up to central terms invisible on endomorphisms. Parallel overlap maps, compatible adjoints and common domains are needed before concluding that the corresponding Hodge and Witten operators glue. Unitary conjugation does preserve the normalized coefficient cost; it does not select those connection or physical-domain data.

### 11.3 The q79 circle extension, not a Nil replacement

Let  $B$  be the selected K3 surface, with classes

$$H^2 = 2, \quad \delta^2 = -4, \quad H \cdot \delta = 0,$$

and primitive  $\delta$ . The topology under discussion is  $X_{q79} = P_\delta \times S_{\text{shared}}^1$ , where the principal circle bundle  $P_\delta \rightarrow B$  has Euler class  $\delta$ . K3 formality and the circle-extension model give the rational CDGA

$$\mathcal{A}_{q79} = H^*(B; \mathbb{Q}) \otimes \Lambda(u, v), \quad |u| = |v| = 1, \quad du = \delta, \quad dv = 0 \quad [12, 17].$$

The source-specific calculation uses the full intersection lattice  $3U \oplus 2E_8(-1)$ , not a fitted list of Betti numbers [54]. There are  $24 \cdot 2 \cdot 2 = 96$  rational basis states. This is not the physical 96-dimensional Dirac carrier merely because the numbers agree. The generator  $v$  records the shared circle topologically, not its connection, charge spectrum or Lorentzian time.

Write  $t$  for the K3 top class, with  $xy = (x \cdot y)t$  in degree four, and put

$$\gamma = \frac{\delta}{\delta^2} = -\frac{1}{4}\delta, \quad x_\perp = x - \frac{\delta \cdot x}{\delta^2}\delta.$$

On the factor without  $v$ , the source projection  $p$  and homotopy  $h$  are

|        |   |                                    |           |     |            |      |
|--------|---|------------------------------------|-----------|-----|------------|------|
| $z$    | 1 | $x$                                | $t$       | $u$ | $ux$       | $ut$ |
| $p(z)$ | 1 | $x_\perp$                          | 0         | 0   | $ux_\perp$ | $ut$ |
| $h(z)$ | 0 | $\frac{\delta \cdot x}{\delta^2}u$ | $u\gamma$ | 0   | 0          | 0    |

and extend by tensoring with the identity on  $\Lambda(v)$ . Here  $p$  is the ambient idempotent onto representatives; factoring it through its image gives the usual projection and inclusion. They obey

$$dh + hd = I - p, \quad p^2 = p, \quad h^2 = ph = hp = 0.$$

This orthogonality is for the indefinite K3 intersection form, not a selected positive HYM inner product.

Since  $\dim \delta^\perp = 21$ , the retained representatives are  $1, v, x, ux, xv, uxv, ut, utv$ , with  $x \in \delta^\perp$ . Thus

$$(b_0, \dots, b_6) = (1, 1, 21, 42, 21, 1, 1), \quad \dim H^*(X_{q79}; \mathbb{Q}) = 88.$$

The smaller algebra generated by  $H, \delta, u, v$  has 16 source states and only eight cohomology states, of ranks  $(1, 1, 1, 2, 1, 1, 1)$ . It is a submodel, not the full compactification. The cyclic pairing is the coefficient of  $tuv$  in a product. It is nondegenerate but not a positive Hilbert norm.

## 11.4 Strict topology does not mean a free physical theory

For these representatives,  $m_2(a, b) = p(ab)$ . A nonzero  $h(ab)$  is proportional to  $u\gamma$ , possibly with  $v$ . Its product with a retained  $\delta^\perp$  class vanishes because  $\gamma \cdot \delta^\perp = 0$ ; its other products project to zero. Both terms of the ternary transfer therefore vanish:

$$m_3 = 0.$$

All homotopy images contain  $u$ , their pairwise products vanish, and another homotopy annihilates a product already containing  $u$ . These facts also kill every binary transfer tree of arity at least four. The source record confirms all  $88^3 = 681472$  triples and gives the structural all-arity argument. Hence, at this topology-only tier,

$$m_1 = 0, \quad m_2 \text{ survives}, \quad m_n = 0 \quad (n \geq 3).$$

Strict transferred operations do not make the displayed representative inclusion multiplicative. For example,

$$p(H) = H, \quad HH = 2t, \quad p(HH) = 0, \quad h(HH) = 2u\gamma.$$

The homotopy carries the difference. Nor does algebraic  $p$  supply a spectral gap, an analytic Hodge projector or a low-energy truncation.

This result identifies where additional structure is needed, not a no-go for physical interactions. It concerns higher operations transferred from this particular rational cup-product DGA. A nonlinear residual with nonzero higher jets, twisted bundle coefficients, Yoneda operations or analytic overlap data is a different source of vertices. The repair-jet mechanism already explained in [section 13](#) remains valid; its physical coefficients must still be evaluated. The full Hull–Strominger source is not proved formal, and no spacetime scattering amplitude vanishes merely because the topology model has  $m_3 = 0$ .

The contrast with the auxiliary Heisenberg model is deliberate:  $dc = ab$  leaves a nonzero ternary operation there; the selected q79 Hirsch contraction does not. The nonminimal CBF response hierarchy is a third object. None should replace either of the others solely because it is finite-dimensional.

## 11.5 How the compiler uses these results

The hidden projective rank-nine module contributes an honestly descending traceless adjoint of rank  $9^2 - 1 = 80$ . This fits the declared  $3 + 8 + 8 + 80 + 3 = 102$  deformation block, but does not evaluate its endpoint arrays. The twisted coefficient model on the relative Jacobian  $J$  and the Hirsch model on  $X_{q79}$  cannot simply be tensored as if they already lived on the same differential-geometric source. A typed realization between those presentations and the metric/domain comparison of the preceding projection section are required.

For a supplied isometry  $U$ , operations transport exactly to its intrinsic range:

$$\widetilde{M}_n(y_1, \dots, y_n) = UM_n(U^\dagger y_1, \dots, U^\dagger y_n).$$

At arity zero this means  $\widetilde{M}_0 = UM_0$ . The unit, cyclic functional and pairings must also be transported to that range. Using an unrelated ambient unit or ambient rank-normalized trace is not the same construction. This is consistent with the corrected embedded-intertwiner and whole-projector results above.

Thus two obligations are resolved at their stated tiers: projective coefficient descent and rational q79 topology transfer. A selected physical coefficient-to-carrier map, common connection and analytic domains, signed action, and independent map to the accepted 27-state algebra remain distinct tasks. No numerical dimension equality closes those maps.

## 12 Auxiliary pencils and the global gerbe problem

Endomorphism descent does not neutralize the original module gerbe. The distinction in [section 11.1](#) leads to a concrete question: can a spectral object be obtained on a suitable cover even when no trivialization exists over the whole family? The eight source records here answer preparatory questions. They do not identify a topological family class with the normalized analytic gerbe value at one physical endpoint.

### 12.1 Fix the spectral class before using a matrix size

Let  $J = S \times E$ , where  $S$  is a K3 surface with ample polarization  $H^2 = 2$ , and  $D_0 = S \times \{p\}$ . For a smooth ample surface  $C_m \in |mH + 3D_0|$ , adjunction and Noether's formula give

$$\begin{aligned} C_m^3 &= 18m^2, & c_2(J) \cdot C_m &= 72, \\ \chi(\mathcal{O}_{C_m}) &= 3m^2 + 6, & q(C_m) &= 1, \\ p_g(C_m) &= 3m^2 + 6, & b_2(C_m) &= 18m^2 + 74. \end{aligned} \quad (\text{D1})$$

Only the cubic product with two K3 factors and one elliptic divisor survives. Weak Lefschetz gives  $b_1 = 2$ . Subtracting the injected ambient  $H^2(J)$ , with Hodge dimensions  $(1, 21, 1)$ , gives

$$(h^{2,0}, h^{1,1}, h^{0,2})_{\text{van}} = (3m^2 + 5, 12m^2 + 41, 3m^2 + 5), \quad \text{rk } V_m = 18m^2 + 51. \quad (\text{D2})$$

Also  $h^0(J, \mathcal{O}(mH + 3D_0)) = 3(m^2 + 2)$ , by K3 and elliptic Riemann–Roch. Hence the projective parameter dimension equals the first and last vanishing Hodge dimensions [\[69\]](#).

| Quantity                       | Auxiliary $m = 3$ | Physical eta9 $m = 9$ |
|--------------------------------|-------------------|-----------------------|
| Section-space dimension        | 33                | 249                   |
| Projective parameter dimension | 32                | 248                   |
| Vanishing Hodge dimensions     | 32, 149, 32       | 248, 1013, 248        |
| Vanishing local-system rank    | 213               | 1509                  |
| Affine homogenization rank     | 214               | 1510                  |

Both covers have degree three over  $S$ ; that degree does not fix the K3 polarization. The 792-node pencil and lattice below concern  $m = 3$  only. The physical eta9 source uses  $m = 9$ , matching its existing compiler. A new physical rank-213 compiler would use the wrong carrier. Equal ranks of a compiler frame and an affine extension are nevertheless only a consistency check, not an isomorphism of filtered differential objects.

### 12.2 Fiberwise triviality need not glue

For  $\mathcal{L} = \mathcal{O}_J(3H + 3D_0)$ , global generation gives

$$0 \rightarrow \mathcal{K} \rightarrow H^0(J, \mathcal{L}) \otimes \mathcal{O}_J \rightarrow \mathcal{L} \rightarrow 0, \quad \text{rk } \mathcal{K} = 32.$$

The universal incidence is  $\mathcal{C} = \mathbb{P}_J(\mathcal{K})$ : over each point of  $J$ , sections vanishing there form a  $\mathbb{P}^{31}$ . With  $q : \mathcal{C} \rightarrow J$  and  $\eta = c_1(\mathcal{O}_q(1))$ , the projective-bundle formula gives

$$H^3(\mathcal{C}; \mathbb{Z}) = q^*H^3(J; \mathbb{Z}) \oplus \eta q^*H^1(J; \mathbb{Z}). \quad (\text{D3})$$

Its rank is  $44 + 2 = 46$ . For the source class  $h = \delta \smile x$ , with  $\delta \cdot H = 0$  and  $0 \neq x \in H^1(E; \mathbb{Z})$ , one therefore has  $q^*h \neq 0$ . Yet  $h|_{\mathcal{C}} = 0$  on every smooth cover: its degree-three pairings

vanish by  $x^2 = 0$  and  $\delta \cdot H = 0$ , with torsion-free cohomology as supplied by the source hypotheses [77].

A trivialization on each fiber need not vary compatibly around parameter loops. Nonzero  $q^*h$  excludes a universal compatible trivialization, hence a universal rank-one twisted line, on the whole incidence. It does not exclude one on a selected cover. Topological DD zero on that cover is also weaker than analytic gerbe zero.

Removing the discriminant does not remove this family class. The source computes

$$\mathrm{im}\{H_D^3(\mathcal{C}; \mathbb{Q}) \rightarrow H^3(\mathcal{C}; \mathbb{Q})\} = \eta q^* H^1(J; \mathbb{Q}), \quad q^* h|_{\mathcal{C}_U} \neq 0. \quad (\text{D4})$$

The boundary double-incidence resolution has four degree-one generators from a cover point  $z$  and its marked singular point  $w$ . The  $z$  generators map to  $792\eta q^* H^1(J)$ ; the  $w$  generators vanish because odd degree cannot integrate against even Chern classes to the real top degree six of  $J_w$ . The full boundary resolution and mixed-Hodge strictness are also required: lower-weight singular gluing cannot enter pure weight-three cohomology [68]. The small Gysin matrix alone would not prove that it describes every boundary contribution. This decides the earlier survival/boundary dichotomy in favor of survival, but does not evaluate the physical eta9 residual.

### 12.3 The horizontal characteristic and its integral typing

Since  $h \smile (3H + 3D_0) = 0$ , the tube construction gives

$$0 \neq \kappa_{h, \mathbb{Q}} \in H^1(U, V_{\mathbb{Q}}), \quad \mathrm{rk} V_{\mathbb{Q}} = 213. \quad (\text{D5})$$

The primitive ambient space has rank  $44 - 2 = 42$ . The universal-hyperplane map is rationally injective; its pencil version detects the class on a Lefschetz pencil [52, 81]. This is horizontal information about incompatible local trivializations, not a nonzero DD class on an individual fiber.

An integral qualification matters. Set  $L_C = H^2(C; \mathbb{Z})$ ,  $A_{\mathbb{Z}} = i^* H^2(J; \mathbb{Z})$  and  $V_{\mathbb{Z}} = A_{\mathbb{Z}}^{\perp}$ . For the source's primitive ambient embedding in the unimodular surface lattice, restriction of the pairing gives

$$L_C/A_{\mathbb{Z}} \simeq V_{\mathbb{Z}}^{\vee}, \quad [V_{\mathbb{Z}}^{\vee} : V_{\mathbb{Z}}] = 2 \cdot 3^{23}. \quad (\text{D6})$$

Thus the integral quotient is not the vanishing lattice itself. An integral trivialization torsor must retain its actual coefficient lattice. A rational edge does not by notation become a canonically normalized class in  $H^1(U, V_{\mathbb{Z}})$ ; that requires a lattice comparison or explicitly declared denominator clearing. Nonzero rationalization proves non-torsion for an integral representative when supplied, but does not construct that lift. Schnell explicitly distinguishes the integral quotient and vanishing lattices [81]. We retain rational nonvanishing, the family obstruction and the following lattice theorem, while qualifying the stronger integral identification in the abbreviated source formulation.

### 12.4 Local double traversal and a common global origin

The auxiliary surface has signature  $(67, 169)$ . For the K3 Gram matrix  $Q$  and polarization vector  $a$ , its ambient form is

$$3 \begin{pmatrix} Q & Qa \\ a^t Q & 0 \end{pmatrix} \simeq 3(Q \oplus \langle -2 \rangle).$$

Its Smith invariants are twenty-two 3s and one 6. The even orthogonal vanishing lattice has signature  $(64, 149)$  and discriminant group  $(\mathbb{Z}/3)^{22} \oplus \mathbb{Z}/6$  [46]. Its roots  $e_i^2 = -2$  generate it integrally and form a monodromy orbit. If all had divisibility two, every Gram entry would be even, forcing  $2^{213} \mid \det V_{\mathbb{Z}}$ ; the determinant actually has 2-adic valuation one. Therefore every root has divisibility one.

For  $T_i(v) = v + (v, e_i)e_i$ , this implies

$$(T_i - I)V_{\mathbb{Z}} = \mathbb{Z}e_i, \quad \lambda_i = a_i e_i \ (a_i \in \mathbb{Z}) \implies [\lambda_i]_{\text{local}} = 0, \quad (I + T_i)\lambda_i = 0. \quad (\text{D7})$$

The local conclusion holds on  $V_{\mathbb{Z}}^{\vee}$  as well: divisibility one gives  $V_{\mathbb{Z}}^{\vee} \cap \mathbb{Q}e_i = \mathbb{Z}e_i$ ; primitivity gives  $(T_i - I)V_{\mathbb{Z}}^{\vee} = \mathbb{Z}e_i$ . The integral comparison qualification therefore does not undo the local root calculation. The selected torsor's meridian must still be identified with that rational vanishing line.

The earlier isolated sign-line model had  $H^1(C_2, \mathbb{Z}_-) = \mathbb{Z}/2$ ; this is not the answer for the full auxiliary lattice. In general,  $(v \mapsto Tv + a)^2$  translates by  $(I + T)a$ , so invariant translations double, not vanish. It is the selected local root calculation, not double traversal alone, that removes the local component here [42].

For a simple global counterexample, let two free generators act linearly by the swap  $T(x, y) = (y, x)$ , with translations 0 and  $(1, -1)$ . Each affine generator squares to the identity and can separately be conjugated integrally to  $T$ . A common origin would require both  $(T - I)b = 0$  and  $(T - I)b = (1, -1)$ , which is impossible. This illustrates, without replacing the geometric proof, why zero local components can coexist with a globally nonzero characteristic. The marking-dependent translations are not independent physical constants.

## 12.5 An explicit pencil, not a physical endpoint

Two integer  $3 \times 11$  section matrices realize the auxiliary existence statement. The basis is the ten cubic K3 monomials and its weight-three coordinate, tensored with three elliptic sections. For  $C(u) = C_{\text{second}} + uC_{\text{smooth}}$ , nine product charts certify smoothness of the base locus modulo 101. All critical points lie in one dense chart, with algebra

$$A_{101} \simeq \mathbb{F}_{101}[u]/(f_{101}), \quad \deg f_{101} = 792, \quad \gcd(f_{101}, f'_{101}) = 1. \quad (\text{D8})$$

Product-minor and ratio-minor elimination and an independent rational-univariate computation give the same polynomial [39]. Equality of algebra length and eliminant degree makes  $u$  separating; squarefreeness gives distinct geometric points, not 792 rational points over the finite field.

Together with the universal degree-792 discriminant and smooth member at infinity, good reduction gives 792 distinct characteristic-zero A1 fibers. Merely finding modular points would not suffice; the finite critical scheme, proper pencil and degree bound matter. The modular nodal anchor at  $u = 0$  also does not assert that the integer member is singular over  $\mathbb{Q}$ . This is an explicit auxiliary witness, not an upstream uniqueness rule or an eta9 physical path. This editorial review reads the recorded argument and arithmetic; it does not rerun the large elimination campaign.

## 12.6 Compress the representation, not the source obligations

For a specified integral lattice  $\Lambda$  and cocycle  $\lambda(gh) = \lambda(g) + \rho(g)\lambda(h)$ , set

$$\tilde{\rho}(g) = \begin{pmatrix} \rho(g) & \lambda(g) \\ 0 & 1 \end{pmatrix}, \quad 0 \rightarrow \Lambda \rightarrow \tilde{\Lambda} \rightarrow \mathbb{Z} \rightarrow 0. \quad (\text{D9})$$

Block multiplication proves the representation law. Changing one common origin gives a coboundary; splitting means vanishing of that class. This works for either correctly specified integral lattice, not only a rational basis [63].

On the auxiliary punctured pencil the rank is  $213 + 1 = 214$ . For a local A1 reflection put  $P_- = (I - T)/2$ . The convention  $\nabla = d - \frac{1}{2}P_- dt/t$  has monodromy  $T$ ; after  $t = s^2$ , its monodromy is trivial on the local cover. The order-two root stack records the sign in its isotropy. This gives local regular boundary charts, not a global splitting; the usual parabolic correspondence requires the appropriate normal-crossing boundary data [4].

A regular-singular extension connection

$$\nabla_{\text{ext}} = \begin{pmatrix} \nabla_{\text{GM}} & \vartheta \\ 0 & d \end{pmatrix}$$

can encode all marked meridian matrices and translations. This packaging does not evaluate  $\vartheta$ , supply an integral marking or normalize the differential-gerbe lift. A complex connection alone does not recover the period lattice. Affine homogenization transfers to eta9; the auxiliary ranks, 792 critical values and divisibility-one proof do not. Its application requires the actual rank-1509 source and comparison with its rank-1510 augmentation.

## 12.7 What would prove an individual analytic zero?

The Euler sequence gives  $H^0(\mathbb{P}^{32}, \Omega^1) = 0$ . For one smooth irreducible divisor of degree  $d > 0$ , the residue sequence also gives  $H^0(\Omega^1(\log \Delta)) = 0$ : the connecting map sends 1 to  $dH$ . Neither is a vector bundle carrying the proposed nonconstant global lift. The value  $c_{32}(\Omega_{\mathbb{P}^{32}}^1) = 33$  is not its zero count. The actual discriminant is singular; the smooth-boundary calculation tests a shortcut, not its resolved logarithmic bundle [42].

The target must keep the normalized gerbe, actual integral image and differential refinement. A quotient such as  $H^2(C, \mathcal{O}_C)/\text{im } H^2(C, \mathbb{Z})$  is not automatically a Hausdorff complex torus. Standard intermediate-Jacobian Neron models concern polarized weight-minus-one variations [82]; they do not apply to this degree-two gerbe quotient just by renaming it a normal function. The filtered/derived comparison and representability argument must be supplied. Likewise an algebraic zero-locus theorem does not prove a nonempty zero [7].

If a correctly typed extended group or torsor object, section and localized zero-section intersection are constructed, then

$$\deg Z_{\text{interior}} = \deg Z_{\text{total}} - \deg Z_{\text{boundary}}. \quad (\text{D10})$$

Positive interior degree forces nonempty support under those intersection hypotheses; it is not obtained from the diagnostic 33. An intrinsic construction may retain nonzero global characteristic and boundary components. Demanding that every class vanish would unnecessarily force the different global-vector route.

The resulting division of work is coherent: the auxiliary family explains why local and global repair differ; the physical eta9 compiler supplies the correct carrier for selected calculation; a normalized analytic comparison, certified global transport and period decision remain necessary for that endpoint. Existing finite-quotient and small-chart results are retained. This synthesis neither sets  $\beta_C = 0$ , supplies  $b_C$ , nor reopens established hidden-carrier existence at its separate tier.

## 13 Higher repair jets and graph shadows

The operator is only the quadratic shadow of nonlinear repair. Higher derivatives of the same cost provide its formal interaction tensors.

Let  $E$  and  $F$  be finite-dimensional inner-product spaces, let  $\Phi : E \rightarrow F$ , and allow a positive residual metric field  $W(a)$ . At a background  $a_*$ , write

$$R_m = D^m \Phi(a_*), \quad W_m = D^m W(a_*), \quad V_n = D^n \mathcal{C}(a_*),$$

where

$$\mathcal{C}(a) = \frac{1}{2} \langle \Phi(a), W(a) \Phi(a) \rangle.$$

**Theorem 13.1** (Complete repair-jet formula). *For every ordered three-color partition  $[n] = A \sqcup B \sqcup C$ , preserving the original argument order within each block,*

$$V_n(v_1, \dots, v_n) = \frac{1}{2} \sum_{A \sqcup B \sqcup C = [n]} \langle R_{|A|}(v_A), W_{|B|}(v_B) R_{|C|}(v_C) \rangle. \quad (13.1)$$

At a zero-defect background,

$$V_2(v, w) = \langle R_1(v), W_0 R_1(w) \rangle.$$

If  $W$  is constant and  $\Phi$  is polynomial of degree  $r$ , then  $V_n = 0$  for  $n > 2r$ .

*Proof.* Apply the multivariable Leibniz rule to the three factors  $\Phi, W, \Phi$ . Each derivative label is assigned exactly once to the left residual, metric, or right residual. Summing over all assignments gives (13.1). If  $R_0 = 0$ , the terms with an empty left or right residual block vanish at order two. Polynomial termination is immediate from the maximum derivative order of the two residual factors.  $\square$

For the Maurer–Cartan residual  $F(a) = da + a^2$  with constant metric, the repair cost therefore has only quadratic, cubic, and quartic vertices. They are not independent knobs: they are derivatives of one residual and one metric.

Once the reduced inverse  $G$  of  $V_2$  is declared, Wick contraction organizes the formal coefficients by graphs [88]. The modal diagrammatics paper proves this graph expansion and its compatible projection theorem [78]. The present theorem adds the missing upstream arrow:

$$(\Phi, W, a_*) \longmapsto (V_2, G, \{V_n\}_{n \geq 3}) \longmapsto \text{formal repair graphs}.$$

This explains why Gaussian and graph structures recur. Every sufficiently smooth local repair process has a quadratic tangent shadow, and every formal quadratic expansion admits Wick organization. What is universal is the grammar. A physical Feynman rule additionally needs a signed action, grading, gauge complex, state, causal prescription, and renormalization.

## 14 Why repair is not automatically action

This section is not a caveat added after the fact. It is a theorem-level separation. Without it, the preceding construction would overclaim exactly where it becomes most physically interesting.

### 14.1 The variational-anchor test

Let  $E, F$  be finite-dimensional real Hilbert spaces and  $\Phi : E \rightarrow F$ . A constant variational anchor is a map  $A : F \rightarrow E^*$  defining the one-form

$$\alpha_a(v) = (A\Phi(a))(v).$$

**Theorem 14.1** (Finite Helmholtz gate). *On a star-shaped domain, a scalar functional  $S_{\text{var}}$  with  $dS_{\text{var}} = \alpha$  exists if and only if  $\alpha$  is closed, equivalently*

$$D(A\Phi)(a) = D(A\Phi)(a)^T \quad (14.1)$$

for every  $a$ . When it exists, one reconstruction is

$$S_{\text{var}}(a) = S_{\text{var}}(0) + \int_0^1 \alpha_{ta}(a) dt. \quad (14.2)$$

*Proof.* Exact one-forms are closed. Conversely, (14.2) is the Poincare radial homotopy on a star-shaped domain; differentiation and  $d\alpha = 0$  give  $dS_{\text{var}} = \alpha$ . The coordinate form of closedness is (14.1).  $\square$

In local field theory the same question becomes the graded formal-adjoint Helmholtz problem on jet bundles, with boundary terms and operator domains retained explicitly [85]. A positive metric on residual space does not supply this anchor.

### 14.2 The universal first-order lift

Even when a direct anchor is absent, introduce a multiplier  $\lambda \in F$  and define

$$S_{\text{mult}}(a, \lambda) = \langle \lambda, \Phi(a) \rangle. \quad (14.3)$$

**Theorem 14.2** (Multiplier lift and normal square). *Every exact closure state  $a_*$  gives a lifted critical point  $(a_*, 0)$ . If  $J = D\Phi(a_*)$ , the Hessian is*

$$\mathcal{D}_J = \begin{bmatrix} 0 & J^\dagger \\ J & 0 \end{bmatrix}, \quad \mathcal{D}_J^2 = \begin{bmatrix} J^\dagger J & 0 \\ 0 & J J^\dagger \end{bmatrix}. \quad (14.4)$$

Moreover,

$$\ker \mathcal{D}_J = \ker J \oplus \ker J^\dagger.$$

*Proof.* Varying (14.3) gives  $\Phi(a) = 0$  and  $D\Phi(a)^\dagger \lambda = 0$ . Differentiating again at  $(a_*, 0)$  gives the off-diagonal block operator. Direct multiplication gives its square and kernel.  $\square$

The multiplier lift is a genuine first-order square root of the repair normal operator. It does not decide whether the multiplier is an ordinary field, response field, antifield, or unphysical auxiliary variable. That typing belongs to a later physical construction, such as a selected BV complex [79].

### 14.3 Information erased by the square

**Proposition 14.3** (Normal-square nonselection). *The positive operator  $J^\dagger J$  does not determine the sign or phase of  $J$ , the signed action Hessian, a causal inverse, or the first-order response. In particular,*

$$J_+ = \text{diag}(2, 3), \quad J_- = \text{diag}(-2, 3)$$

*have the same normal square  $\text{diag}(4, 9)$ , but their inverse responses differ in the sign of the first channel.*

If a signed self-adjoint action Hessian  $H_{\text{var}}$  is identified with  $J$ , then the repair Hessian is  $H_{\text{var}}^2$  in the simplest equal-metric case. Squaring erases the Morse index. A saddle direction of the physical action becomes a positive repair direction.

The interaction vertices also differ. If

$$S_{\text{var}}(x) = \frac{m}{2}x^2 + \frac{g}{6}x^3, \quad \Phi(x) = S'_{\text{var}}(x) = mx + \frac{g}{2}x^2,$$

then

$$S_{\text{rep}}(x) = \frac{1}{2}\Phi(x)^2 = \frac{m^2}{2}x^2 + \frac{mg}{2}x^3 + \frac{g^2}{8}x^4.$$

The signed-action rows  $(V_2, V_3, V_4) = (m, g, 0)$  become the repair rows  $(m^2, 3mg, 3g^2)$ . Repair graphs are therefore not physical Feynman graphs unless a separate theorem identifies the physical perturbative datum with the repair functional.

### 14.4 When a signed action does exist

Let  $\mathcal{A}$  be a differential graded associative algebra with a degree-three cyclic trace  $\tau$  satisfying

$$\tau(dx) = 0, \quad \tau(xy) = (-1)^{|x||y|}\tau(yx).$$

For  $a \in \mathcal{A}^1$ , define

$$S_{\text{MC}}(a) = \frac{1}{2}\tau(a da) + \frac{1}{3}\tau(a^3). \tag{14.5}$$

**Theorem 14.4** (Cyclic Maurer–Cartan action). *For every degree-one variation  $u$ ,*

$$DS_{\text{MC}}(a)[u] = \tau(u(da + a^2)). \tag{14.6}$$

*If the pairing between degrees one and two is nondegenerate, the critical locus is exactly the Maurer–Cartan locus. The action is infinitesimally gauge invariant under  $\delta_\epsilon a = d\epsilon + [a, \epsilon]$ . It descends through even projective overlaps because the scalar cocycle cancels under conjugation and the cyclic trace is conjugation invariant.*

*Proof.* Differentiate the quadratic term. The Stokes identity moves  $d$  from the variation to  $a$ , making the two contributions equal. Cyclicity makes the three cubic contributions equal. This gives (14.6). Pairing a gauge variation with the residual and integrating by parts yields the Bianchi identity. On overlaps, each monomial is conjugated by the same even transition, and the trace removes that conjugation.  $\square$

On a compact complex threefold with holomorphic volume form  $\Omega$ , the candidate trace is

$$\tau(x) = \int_X \Omega \wedge \text{Str}(x),$$

and (14.5) becomes holomorphic Chern–Simons theory [90, 11]. This supplies a signed action for the integrability lane. It does not automatically supply nondegenerate cyclic pairings for the HYM, balanced, anomaly, coframe, and Lorentzian lanes.

## 15 From anomaly data to a constrained repair problem

There is a useful intermediate answer to the action question. The anomaly equation need not be an unrelated extra row appended to a bundle complex. It can help define the complex itself. Three existing source results make this precise: a string-algebroid constructor, a Bott–Chern lift of bundle deformations, and a moment-map interpretation of the remaining equations [57, 53, 71]. Their order matters. Constructing the complex does not solve its moment-map equations, and solving a form equation over real or complex coefficients does not select an integral flux sector.

### 15.1 The rank-102 carrier acquires structure

Let  $P$  be the holomorphic product principal bundle for the tangent, visible and projective hidden factors, and let  $c$  be a specified nondegenerate invariant bilinear pairing on its Lie algebra. Given a compatible connection  $\theta$  and a form  $H \in \Omega^{3,0} \oplus \Omega^{2,1}$  satisfying

$$F_\theta^{0,2} = 0, \quad dH + c(F_\theta \wedge F_\theta) = 0, \quad (\text{S1})$$

the standard string-algebroid construction supplies the holomorphic Courant structure, including its anchor, pairing and bracket [15, Proposition 2.4]. It fits into

$$0 \longrightarrow T^*X \longrightarrow Q \longrightarrow A_P \longrightarrow 0, \quad 0 \longrightarrow \text{ad } P \longrightarrow A_P \longrightarrow TX \longrightarrow 0.$$

Its smooth splitting has complex rank  $3 + 8 + 8 + 80 + 3 = 102$ . The finite central quotient of the hidden group leaves its rank-80 adjoint honest. It does not trivialize the hidden fundamental gerbe. Thus rank matching identifies the carrier type; the anomaly triple supplies its operations. Once that triple is given, a bracket is not another free choice.

The q79 promotion uses more than an invariant-form computation. The full real map  $H_{BC}^{2,2}(X) \rightarrow H_{\text{dR}}^4(X; \mathbb{R})$  is injective on the specified q79 geometry. For supplied holomorphic product bundles with the required total characteristic cancellation, choose Chern connections. Their real, closed, pure-type form  $\Xi = c(F_\theta \wedge F_\theta)$  is de Rham exact. Consequently one may write, in the source’s convention,

$$\Xi = 2i\partial\bar{\partial}\tau, \quad H = 2i\partial\tau, \quad dH = -\Xi. \quad (\text{S2})$$

The sign follows from  $\bar{\partial}\partial = -\partial\bar{\partial}$ . This is a conditional existence construction for the complex anomaly data, not a numerical solution of the HYM equations. Changing the primitive can change the string extension class. Nor is this complex truncated form automatically the real torsion  $H_{\mathbb{R}} = d^c\omega$ : reality, trace coefficients, integral refinements and the tangent-connection convention must still match. Existing existential hidden-bundle/HYM results are not reopened by these additional requirements on a common physical source.

A small exact witness checks the mechanism, not the continuum existence premises. Take  $\partial\tau = h$ ,  $\bar{\partial}\tau = k$ ,  $\partial k = v$ ,  $\bar{\partial}h = -v$ , with all further maps zero. Three closed curvature labels with squares  $2v$ , weighted by  $(3/2, -1/2, -1/2)$ , give  $\Xi = v$ . Then  $H = h$  solves  $dH + \Xi = 0$ . This rational form complex verifies the transgression sign and product-gauge accounting; it is not a selected curvature table.

### 15.2 What the Bott–Chern lift actually lifts

The form lane remains visible in the deformation algebra. Write  $\mathfrak{g}^k = \Omega^{0,k}(\text{ad } P)$  and  $\mathfrak{F}^k = \bigoplus_{j=0}^k \Omega^{j+2,k-j}$ , omitting types outside the dimension of  $X$ . In nonnegative degrees the

string differential graded Lie algebra  $\mathfrak{L}_Q$  is an extension of  $\mathfrak{g}$  by this abelian form complex. In the constructor's convention,

$$d_Q(a, b) = (\bar{\partial}_\theta a, db - 2c(\partial_\theta a \wedge F_\theta)).$$

The bracket also has a quadratic form component. In particular, for a bundle Maurer–Cartan solution  $\bar{\partial}_\theta a + \frac{1}{2}[a, a] = 0$ , define

$$\Delta(a) = 2c(\partial_\theta a \wedge F_\theta) + c(\partial_\theta a \wedge \partial_\theta a). \quad (\text{S3})$$

The full equation is  $db = \Delta(a)$ , and the deformed anomaly triple is  $(\theta + a, H - b)$ , not  $(\theta + a, H + b)$ . On the fixed smooth bundle,  $\Delta(a)$  is closed, pure  $(2, 2)$  and de Rham exact. Complexifying the full q79 Bott–Chern injectivity result gives  $\Delta(a) = 2i\partial\bar{\partial}\tau_a$ ; hence  $b = -2i\partial\tau_a$  solves it exactly. The connection  $\theta + a$  need not be the Chern connection of the old Hermitian metric.

An auxiliary Hermitian elliptic solution operator  $\mathcal{K}_{BC}$  can select a primitive on this exact-form subspace. It defines a section of Maurer–Cartan *sets*,

$$s_{\mathcal{K}}(a) = (a, \mathcal{K}_{BC}\Delta(a)), \quad \pi s_{\mathcal{K}}(a) = a. \quad (\text{S4})$$

For each  $a$ , its unquotiented fiber is affine over the closed degree-one form lane. The auxiliary metric chooses a representative, not a physical flux. The section is generally nonlinear, and gauge-equivariance, integral refinement or an  $L_\infty$  equivalence do not follow from  $\pi s = 1$ . The representable product-adjoint subcomplex embeds as a Lie dg algebra in the commutator algebra of cohesive endomorphisms. It is not an associative dg subalgebra: traceless matrices are closed under commutators, but  $E_{12}E_{21} = E_{11}$  has nonzero trace. Arbitrary off-diagonal or higher cohesive deformations are not covered by this lift.

The finite source example makes the distinction tangible. In degree one use  $(e, b, z)$ , in degree two  $(c, v)$ , and set

$$de = c - 2v, \quad db = v, \quad dz = 0, \quad [e, e] = 2c - 2v,$$

with all other brackets and degree-two differentials zero. Its Maurer–Cartan residual at  $xe + yb + \lambda z$  is

$$(x + x^2)c + (y - 2x - x^2)v.$$

Thus  $s(x) = (x, 2x + x^2, 0)$  lifts precisely the two base roots  $x = 0, -1$ , and  $\lambda z$  remains free. The polynomial extension obeys  $s(2) - 2s(1) = 2b$ , so it is not a linear splitting. A source section has been constructed; uniqueness of a complete physical deformation has not.

### 15.3 One moment map, with distinct volume-form roles

On a compact real form, the integrable positive horizontal-lift space has the Calabi moment-map description. For fixed positive volume  $\mu$ , put  $f_\omega = \frac{1}{2} \log((\omega^n/n!)/\mu)$ . Its zero locus consists of

$$F_\theta \wedge \omega^{n-1} = 0, \quad d(e^{-\ell f_\omega} \omega^{n-1}) = 0,$$

together with the integrability equations  $F^{0,2} = 0$  and  $dd^c \omega + c(F \wedge F) = 0$ . For  $n = 3, \ell = 1$ , and the holomorphic volume convention,  $e^{-f_\omega} = \|\Omega_{\text{hol}}\|_\omega$ . This recovers the Hull–Strominger equations [16, Proposition 5.14 and Corollary 5.15]. For the MTT product group, the three HYM rows and the balanced row are components of one moment map; the integrability and

anomaly equations instead define its configuration stratum. Matching the selected physical trace and real torsion conventions is required for this interpretation.

The normalization row has a different role again. If  $\mu_{\Omega_{\text{hol}}}$  is the positive volume of the holomorphic form, define

$$\Omega_{\text{unit}} = e^{f\omega} \Omega_{\text{hol}}, \quad \mu_{\Omega_{\text{unit}}} = e^{2f\omega} \mu_{\Omega_{\text{hol}}} = \omega^3/3!. \quad (\text{S5})$$

This is a coordinate identity. In general  $d\Omega_{\text{unit}} = e^{f\omega} df_{\omega} \wedge \Omega_{\text{hol}} \neq 0$ . Requiring the same form to be both holomorphic and unit normalized adds a constant-norm restriction. The conformally balanced equation must use  $\Omega_{\text{hol}}$ ; replacing it by the unit representative would wrongly force  $d\omega^2 = 0$ . Thus the two uses of the symbol  $\Omega$  in an ambient residual compiler must be separated before physical use.

#### 15.4 Restrict the Hessian, not the ambient proof

Suppose an actual stratum inclusion  $\iota : \mathcal{Z} \rightarrow \mathcal{C}$  implements the constructor, normalization chart and same-complex identification just described. At an exact solution, let  $j = D\iota$ . If  $\mathcal{A} \circ \iota = \mathcal{N} \circ \iota = 0$ , then

$$DAj = DNj = 0, \quad \text{Hess}(\mathcal{C} \circ \iota) = j^\dagger D\Phi^\dagger W D\Phi j. \quad (\text{S6})$$

Here  $W$  is any chosen positive residual metric. All terms involving the vanished rows, including cross terms, vanish on this tangent space. The proof is the chain rule at  $DC = 0$ , not cancellation of positive terms in the ambient Hessian. The normal and obstruction complex is still needed. Gauge/diffeomorphism slices concern the quotient analysis; they are not additional physical residual equations.

For example, take an ambient  $\mathbb{R}^{11}$  with coordinates  $(\phi, a, \mu_1, \dots, \mu_4, n, g_1, \dots, g_4)$ . Let  $J = (I_7 \ 0)$ . Its Euclidean Gram matrix has rank seven. Inclusion of  $(\mu_1, \dots, \mu_4, g_1, \dots, g_4)$ , setting  $\phi = a = n = 0$ , pulls it back to  $\text{diag}(I_4, 0_4)$ , of rank four. The three removed directions are transverse, not three newly discovered physical gauge modes. This conditional restriction does not contradict the ambient 25-block support in [section 19](#), and is not proof of a reducing or isometric spectral intertwiner. Without the inclusion and convention match, deleting the anomaly row remains unjustified.

#### 15.5 The action bridge and its positivity boundary

With convention  $d\langle \mu, \xi \rangle = \iota_{\rho_\xi} \varpi$  and  $g(u, v) = \varpi(u, Iv)$ , one has  $\text{grad}_g \langle \mu, \xi \rangle = I\rho_\xi$  where the metric is nondegenerate. This ties the moment-map components to infinitesimal complexified gauge motion. The rational witness on  $\mathbb{R}_x^4 \oplus \mathbb{R}_y^4$  has standard  $\varpi, I, g = I_8, \rho_i = -\partial_{y_i}$  and  $\mu_i = x_i$ . Its orbit Jacobian is  $I_4$ . It illustrates a variational anchor but does not establish positivity in the geometric moduli problem.

In particular, the cited Calabi construction is generally *pseudo-Kahler*, possibly degenerate, not automatically a positive repair metric. A positive horizontal lift means that its underlying Hermitian form is positive; it does not give positivity of every field-space metric. The anomaly pairing has signed factor weights. Nondegeneracy, positive directions, convexity and coercive estimates need their own hypotheses; at degeneracy an inverse-metric gradient is not available. Neither a local orbit-potential argument nor the finite positive witness establishes global minimization or a Lorentzian physical action.

The later cyclic-cotangent result is already incorporated in the action discussion: the uncompleted string form lane does not automatically inherit a nondegenerate cyclic trace,

whereas a shifted cotangent completion supplies dual variables. Shared-line compatibility on the neutral product-adjoint sector likewise does not identify every fundamental charged field. These three imports therefore remove independent bracket and form-lift searches on their stated domain, organize the residual roles, and feed the existing BV construction. They do not select the common physical endpoint, the remaining flux data or the four-dimensional action.

## 16 From a cotangent lift to a four-dimensional BV theory

The action boundary has a constructive continuation. A multiplier pairs an equation with a dual variable; Batalin–Vilkovisky (BV) geometry organizes fields, ghosts and antifields with an odd symplectic pairing. Antifields are dual bookkeeping variables for the gauge equations, not additional observed particles. With consistent ghost degrees, the classical master equation  $\{S, S\} = 0$  expresses compatibility of the action and gauge differential. A volume element, integration prescription and quantum master equation are further data [83]. The following two source results explain what this construction preserves under reduction [35, 50].

### 16.1 Retain fields and their correct duals together

Let  $(L, d_L)$  retract onto  $(H, d_H)$  by cochain maps  $p, i$  and a degree-minus-one map  $h$ , with

$$pi = 1, \quad d_L h + h d_L = 1 - ip, \quad h^2 = ph = hi = 0.$$

Assume finite-dimensional complexes, or locally perfect complexes with a specified density-valued dual and nondegenerate evaluation. A functional analytic realization must additionally specify domains and continuous duals. Write  $p^!, i^!$  for the graded density transposes. In the cyclic DGLA convention used by the internal source, the cotangent complexes and maps are

$$\begin{aligned} \widehat{L} &= L \oplus L^![-3], & \widehat{H} &= H \oplus H^![-3], \\ \widehat{i} &= i \oplus p^!, & \widehat{p} &= p \oplus i^!. \end{aligned} \tag{16.1}$$

The suspension to field coordinates gives the degree-minus-one BV pairing; the bracket notation in (16.1) is not itself a spacetime dimension. Transpose the homotopy with the same graded convention. For the ordinary block-matrix check, the differential and homotopy are

$$\widehat{d} = \text{diag}(d_L, -d_L^T), \quad \widehat{h} = \text{diag}(h, -h^T).$$

Then

$$\widehat{p}\widehat{i} = 1, \quad \widehat{d}\widehat{h} + \widehat{h}\widehat{d} = 1 - \widehat{i}\widehat{p}, \quad \widehat{i}^*\omega_L = \omega_H.$$

The pairing identity follows from  $(p^! \eta)(iy) = \eta(piy) = \eta(y)$ . Thus it is the *inclusion of the retained field–dual sector*, not the projection on the entire upper space, that preserves the nondegenerate pairing.

For example, retain  $r_1, r_2$  from a complex with basis  $(r_1, r_2, u, v)$ ,  $du = v$ , and  $h(v) = u$ . The cotangent lift has eight coordinates above and four below. A discarded pair  $u, u^*$  has pairing one above and zero after projection. This elementary example explains why deleting modes is not automatically a symplectic equivalence. Restricting to the retained symplectic sector is exact; integrating the complement instead requires a suitable BV pushforward. Such effective-action constructions are established mathematics [10]; the existing downstream q79 results apply after their four-dimensional BV complex and compatible split have been supplied.

A cochain contraction alone does not transport an arbitrary nonlinear action. If the retained nonlinear maps obey  $Q_L i = i Q_H$ , then

$$S_L(\widehat{i}(x, \eta)) = \langle p^! \eta, Q_L(ix) \rangle = \langle \eta, Q_H(x) \rangle.$$

For a BV interpretation  $Q$  must have the appropriate degree and satisfy  $Q^2 = 0$ ; an arbitrary residual only defines a multiplier action. With non-strict transfer, all required higher brackets and the compatible cyclic map must be retained. Neither a bare projection nor a truncated residual proves that action identity.

## 16.2 Why the field-only action cannot be obtained by relabeling

The internal multiplier has  $S_{\text{cot}}(x, 0) = 0$ . A supplied four-dimensional BV theory generally has

$$S_{\text{BV}}(z, z^*) = S_0(z) + \sum_i z_i^* Q^i(z), \quad S_{\text{BV}}(z, 0) = S_0(z) \neq 0.$$

Consequently a zero-section-preserving pullback cannot identify these actions without a field-only upper term that reduces to  $S_0$ . This is a restricted obstruction, not a theorem against every canonical transformation: a transformation mixing fields and antifields would need its own generating data and proof.

The cyclic integrability action in the preceding section supplies one signed field-only functional. It is not thereby the complete Standard Model action on a Lorentzian four-manifold. In particular, matching the abstract cotangent degree does not identify internal adjoint/form fields with gauge potentials, charged fermions, ghosts and the Higgs. Their representations, density, real structure, statistics and operator domains must be carried by the same reduction.

## 16.3 The exact unit-orientation construction

There is nevertheless a simple exact reduction mechanism. Use the rational q79 algebra of Section 11.3, and let  $\nu = utv$  be its degree-six top class, with  $\tau(\nu) = 1$ . The two-dimensional subalgebra

$$R = \text{span}\{1, \nu\}, \quad 1\nu = \nu 1 = \nu, \quad \nu^2 = 0$$

has nondegenerate cyclic pairing

$$B_R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

These are two complementary *internal profiles*, not two physical fields or two universes. The raw topological trace has  $\tau(1) = 0$ . A normalized density is therefore essential for a field-only term:  $\tau(\nu S_0) = S_0$ . Choosing an algebraic map  $\star 1 = \nu$ ,  $\star \nu = 1$  gives  $B_R \star = I_2$ , but does not select the physical Hodge star or volume. For a Riemannian metric with volume  $V$  and  $\nu = \text{vol}/V$ , instead  $\star 1 = V\nu$ ,  $\star \nu = 1/V$ ; the pairing matrix is  $\text{diag}(V, 1/V)$ . Topological normalization does not determine  $V$ .

Now supply an antifield-linear classical BV action with  $Q^2 = 0$  and  $Q(S_0) = 0$ , and assign profiles

$$z_i \mapsto z_i \otimes 1, \quad z_i^* \mapsto z_i^* \otimes \nu.$$

Declare the degree-minus-six fiber-integration shift, as well as the field/ghost gradings. Then both the action and odd pairing reduce exactly:

$$\tau \left( \nu S_0(z \otimes 1) + \sum_i (z_i^* \otimes \nu) Q^i(z \otimes 1) \right) = S_{\text{BV}}(z, z^*), \quad (16.2)$$

$$\omega_{\text{red}}(z \otimes 1, z^* \otimes \nu) = \omega_4(z, z^*). \quad (16.3)$$

Without the degree shift these equations establish coefficient agreement, not agreement of BV degrees. The restriction to antifield-linear actions also matters: a term with two such antifield profiles would contain  $\nu^2 = 0$ . General open gauge algebras require additional structure.

A small example has even fields  $x, z$ , odd ghost  $c$ , and  $Qx = c$ ,  $Qc = Qz = 0$ . Choose

$$S_0(z) = \frac{1}{2}z^2 + \frac{1}{3}z^3, \quad S_{\text{BV}} = S_0 + x^*c.$$

Here  $Q^2 = 0$  and  $Q(S_0) = 0$ , so the classical master equation holds. Both terms return unchanged under (16.2). The ghost is a Grassmann variable, not a freely substituted ordinary number; rational coefficient checks are not a replacement for that grading.

The construction has the precise direction  $\text{Red} \circ \text{Lift} = 1$ . It reconstructs a theory already specified below and demonstrates that the reduction rules can fit together. It does not prove that the selected upper MTT source is this lift, or derive the chosen coefficients of  $S_0$ .

#### 16.4 A consistent sector is not all low-energy modes

Tensoring the full internal contraction by an external cochain complex  $(E, d_E)$  is exact. The total differential and homotopy are

$$D(e \otimes a) = d_E e \otimes a + (-1)^{|e|} e \otimes d_A a, \quad K(e \otimes a) = (-1)^{|e|} e \otimes h(a).$$

With  $I = 1 \otimes i$ ,  $P = 1 \otimes p$ , the signs cancel the mixed terms and give  $DK + KD = 1 - IP$  and the normalized side conditions. For the archived three-state external example this is a 288-to-264 contraction, retaining all 88 internal cohomology profiles.

By contrast, the orientation choice keeps only two of them. The cyclic pairing splits

$$H(A) = R \oplus R^\perp, \quad \dim R^\perp = 86,$$

and its restriction to  $R^\perp$  remains nondegenerate. These classes cannot be contracted away by the bare topology differential. Nor need  $R^\perp$  be an ideal: the degree-one class  $v$  and degree-five class  $ut$  lie in the complement but  $v(ut) = -\nu$ . Thus the two-profile projection is not an algebra quotient of the full cohomology.

A consistent classical restriction is still possible when the full Hamiltonian vector field is tangent to the retained symplectic sector. Cyclicity, orthogonality and preservation by all source operations are sufficient for the usual cyclic construction; a newly added field-only term must satisfy the same tangency condition. This is stronger than having a subalgebra. It is weaker than proving that the discarded modes are absent or massive. A physical bundle/flux/Hessian operator would have to constrain or lift the extra profiles and certify the relevant energy scale. The number 86 counts bare rational cohomology classes, not 86 predicted particles.

#### 16.5 The topology-only lifting question is already decided

A stronger follow-on theorem sharpens the preceding boundary [58]. On the *unchanged graded* 88-dimensional carrier,

$$\chi(H) = 1 - 1 + 21 - 42 + 21 - 1 + 1 = 0, \quad \chi(R) = 1 + 1 = 2.$$

Euler characteristic is invariant under replacing a finite cochain complex by its cohomology. Hence no degree-one square-zero differential on this carrier can leave exactly  $R$  as cohomology. If it preserves  $H = R \oplus R^\perp$  and is zero on  $R$ , then

$$\chi(R^\perp) = -2, \quad \dim H(R^\perp, d_{\text{eff}}) \geq 2.$$

Thus at most 84 of the 86 complement profiles can be lifted within this specified cochain model. This is not an unsolved numerical search.

The bound is attained, including cyclic compatibility. Choose rational Poincare-dual bases  $b_i, b_i^\vee$  in degrees two and four and symplectic pairs  $w_i, z_i$  in degree three,  $0 \leq i \leq 20$ . Set  $\langle b_i, b_j^\vee \rangle = \langle w_i, z_j \rangle = \delta_{ij}$ ; let  $c_1 = v$ ,  $c_5 = ut$ , so  $\langle c_1, c_5 \rangle = -1$  in the present orientation convention. Define the only nonzero arrows by

$$\begin{aligned} d_{\text{eff}}c_1 &= b_0, & d_{\text{eff}}b_i &= w_{i-1} \quad (1 \leq i \leq 20), \\ d_{\text{eff}}z_j &= -b_{j+1}^\vee \quad (0 \leq j \leq 19), & d_{\text{eff}}b_0^\vee &= -c_5. \end{aligned}$$

Each image has zero outgoing differential. The paired arrows give  $\langle da, b \rangle + (-1)^{|a|} \langle a, db \rangle = 0$ . The degreewise differential ranks and cohomology ranks are respectively

$$(0, 1, 20, 20, 1, 0, 0), \quad (1, 0, 0, 2, 0, 0, 1).$$

Declaring this adapted basis orthonormal yields  $\Delta_{\text{aux}} = d^\dagger d + dd^\dagger$  with four zero eigenvalues and 84 eigenvalues equal to one. The extra survivors are  $w_{20}, z_{20}$ . This is a cyclic cochain witness; it is not asserted to be a derivation of the original cup product. The metric and unit gap are auxiliary choices, not an HYM metric or a physical mass prediction.

This theorem rules out only the unchanged-carrier differential route. Bundle-valued coefficients, a larger BV complex, a derived gauge quotient, or an independently sourced noncochain mass Hessian change its premises. It does not predict two extra particles. Nor is the topology count 88 the  $8 + 80$  gauge-adjoint fiber count, despite the numerical coincidence. The rank-102 deformation block and the index-bearing matter complex must remain separately typed.

## 16.6 Where physical representations must enter

If a group acts trivially on the topology factor, then

$$(V \otimes A)_\chi = V_\chi \otimes A.$$

Tensoring changes multiplicities, not character support. This elementary obstruction does not identify the shared differential line with hypercharge, and does not say that every adjoint is neutral under every subgroup. An adjoint of a larger group can acquire nonzero weights on restriction. The no-go concerns a *specified action that is trivial on the source*. Charged/chiral modes may instead enter through bundle-valued cohomology, off-diagonal Hom sectors or an index-bearing Dirac complex.

The selected physical comparison must therefore supply one compatible chain: an externalized source and field-only action; its charged fields and duals; density, degree and real-structure maps; a BV-compatible low-energy contraction; Lorentzian gauge-fixed domains; and, for a quantum claim, the measure, integration cycle and master-equation transport. Sections 11.2 and 11.5 still govern the global pairing and normalization. The exact two-profile construction removes the need to invent abstract product reduction. It neither replaces those physical inputs nor reopens the accepted downstream BV results at their declared scope.

## 17 From background repair to a chiral matter test

The preceding topology obstruction makes bundle-valued matter a natural next construction, but a background deformation and a particle field do not have the same job. Four source results connect the first-order index, associated representation, spectral localization and mirror test [38, 37, 55, 49]. Their contribution here is an ordered calculation, not a new derivation of the already-established finite Standard Model representation.

### 17.1 Keep the grading before taking the square

Let  $D_+ : \mathcal{H}_+ \rightarrow \mathcal{H}_-$  be a closed densely defined Fredholm operator with its Hilbert adjoint. With their natural domains,

$$H_+ = D_+^\dagger D_+, \quad H_- = D_+ D_+^\dagger, \quad \text{ind } D_+ = \dim \ker H_+ - \dim \ker H_-.$$

The nonzero spectra pair; the kernels need not. Replacing  $D_+$  by its adjoint reverses the index and exchanges the two positive blocks. Consequently the *ungraded* direct sum  $H_+ \oplus H_-$  does not determine the sign of chirality. This is not a claim that a positive operator loses its index when its grading has actually been retained.

For an exact finite example take  $D_+ = (1, 0, 0, 0) : \mathbb{Q}^4 \rightarrow \mathbb{Q}$ . Its index is 3; the adjoint has index  $-3$ . The full normal matrices have diagonals

$$(1, 0, 0, 0, 1), \quad (1, 1, 0, 0, 0),$$

related by a permutation. With  $\Gamma = +1$  on the source and  $-1$  on the target, however,

$$\text{Tr}(\Gamma f(H_+ \oplus H_-)) = \text{ind } D_+, \quad f(0) = 1.$$

This identity holds for every function on the finite spectrum. In an infinite-dimensional realization the trace must exist; for heat kernels this is the graded heat-trace/index mechanism [22, 1]. Compact ellipticity supplies the usual trace-class setting. A general noncompact Fredholm operator alone does not.

If a finite abelian unitary group action commutes with  $D_+$ , the same argument works in every character block, giving

$$k_{+, \chi} + k_{-, \chi} \geq |I_\chi|, \quad I_\chi = k_{+, \chi} - k_{-, \chi}.$$

An inverse-character pairing requires the *equivariant* duality data. Serre duality pairs  $F$  with  $F^\vee \otimes K_X$ ;  $K_X \simeq \mathcal{O}_X$  as a line bundle does not by itself identify their group actions. For an equivariantly self-dual, dual-complete carrier with an invariant holomorphic volume form, the familiar relation is  $I_{\chi^{-1}} = -I_\chi$ . If the canonical line has a nontrivial character, that character must be included in the pairing. The source's finite inverse-character witness uses the invariant-volume case; it does not select the physical q79 character assignment.

On a compact complex threefold with  $c_1(TX) = c_1(F) = 0$ , Hirzebruch–Riemann–Roch gives

$$\chi(X, F) = \int_X \text{ch}_3(F) = \frac{1}{2} \int_X c_3(F).$$

Thus a valid bundle with  $|\int_X c_3(F)| = 6$  has net index magnitude three. This is a difference of multiplicities, not their separate values. By duality  $\chi(X, \text{End}_0 F) = 0$  when  $K_X$  is trivial; the tangent and cotangent contributions also cancel. These facts explain why the rank-102 background deformation carrier is not itself the associated chiral matter complex. Index zero does not mean invertibility.

## 17.2 A representation projector is not a zero-mode count

The existing representation construction supplies  $27 = 16_1 \oplus 10_{-2} \oplus 1_4$ . The 16 here includes the neutral right-handed-neutrino slot of the adopted carrier; it is not a claim that the minimal Standard Model requires this additional field. Let  $Q_\psi$  have the three indicated eigenvalues. Its polynomial projectors are

$$P_{16} = -\frac{(Q_\psi + 2I)(Q_\psi - 4I)}{9},$$

$$P_{10} = \frac{(Q_\psi - I)(Q_\psi - 4I)}{18}, \quad P_1 = \frac{(Q_\psi - I)(Q_\psi + 2I)}{18}.$$

They sum to the identity, are pairwise orthogonal, and have ranks 16, 10, 1. If the associated differential has the form  $D_V \otimes I_{27}$ , or commutes with a parallel  $Q_\psi$ , they are chain projectors. A merely pointwise branching is insufficient when the differential mixes those sectors. The choice of  $Q_\psi$  here does not establish a surviving physical current or require a Wilson line in the independent direct MTT gauge-group construction.

Write  $k_+ - k_- = 3$  and  $m = k_-$ . The associated 16 and conjugate lanes then contain

$$16(k_+ + k_-) = 48 + 32m$$

component zero modes. The net count is 48, but the total is 48 only if the opposite kernel vanishes. Each mirror pair adds 32 components without changing the index. The accepted 48-dimensional representation and anomaly calculation remain valid at their original tier; this condition asks whether one particular geometric operator realizes exactly that spectrum.

## 17.3 What a one-doublet lifting witness does establish

Under the three-copy, no-mirror premise, the remaining  $3(10 \oplus 1)$  sector has dimension 33. In each copy its Standard Model representations have dimensions 3, 3, 2, 2, 1, denoted  $D, D^c, H_u, H_d, S$ , with hypercharges  $-1/3, 1/3, 1/2, -1/2, 0$ . A complex-linear, gauge-equivariant Hermitian normal acts through the multiplicity spaces:

$$H_{\text{ex}} = (M_D \otimes I_3) \oplus (M_{D^c} \otimes I_3) \oplus (M_{H_u} \otimes I_2) \oplus (M_{H_d} \otimes I_2) \oplus M_S.$$

Each  $M_s$  is a positive semidefinite  $3 \times 3$  matrix. Ranks  $(3, 3, 2, 3, 3)$  give

$$\text{rank } H_{\text{ex}} = 9 + 9 + 4 + 6 + 3 = 31, \quad \dim \ker H_{\text{ex}} = 2.$$

The exact witness takes all blocks equal to  $I_3$ , except  $M_{H_u} = \text{diag}(0, 1, 1)$ . Its kernel carries one positive-hypercharge weak doublet. In contrast,  $m_H I_3$  has either zero or three family-kernel directions, never one.

This is an equivariant operator feasibility result. It neither selects a physical mass scale nor turns a fermionic matter component into a bosonic Higgs. Field statistics, conjugation and the signed action must supply that interpretation. Nor may every such positive normal be identified with an arbitrary conjugate-pair mass term: a single square  $3 \times 3$  map  $B$  pairing two conjugate families has normals  $B^\dagger B$  and  $BB^\dagger$  of equal rank, so cannot alone realize the unequal doublet ranks 2 and 3. Additional bosonic terms, symmetry breaking or another explicitly typed field construction must justify the proposed physical realization. The unit positive gap of the witness is only its chosen normalization.

## 17.4 Localizing the family question on a curve

The family multiplicity can be tested before normalizing harmonic wavefunctions. For this route take a holomorphic principal elliptic bundle  $\pi : X \rightarrow B$ , with translation structure group and relative Jacobian  $J = B \times E^\vee$ . The zero section is  $\sigma : B \rightarrow J$ , with image  $D_0$ . Use the normalized BHT analytic gerbe  $\alpha$ , its specified zero-section trivialization  $t_0$ , and the dual normalized universal kernel  $\mathcal{P}^\vee$ . The underlying twisted transform is established in [6]; spectral cohomology localization also has the classical bundle-theoretic precedent [14].

Let  $i : C \hookrightarrow J$  be a smooth surface finite flat over  $B$  carrying an  $\alpha|_C$ -twisted line  $\mathcal{L}_\alpha$ . Assume a smooth transverse intersection  $\Sigma = C \cap D_0$ . The trivialization  $t_0$  makes the restriction an ordinary line  $M$  on  $\Sigma$ . In the normalized convention of the source result,

$$R\pi_* V = s_* M[-1], \quad R\Gamma(X, V) = R\Gamma(\Sigma, M)[-1], \quad (17.1)$$

where  $V = \Psi(i_* \mathcal{L}_\alpha)$  and  $s : \Sigma \rightarrow B$ .

The shift has content. Locally the elliptic transform of the normalized Poincare kernel is  $\mathcal{O}_0[-1]$ . A derived fiber at the origin has two adjacent classes because the skyscraper's Cartier resolution acquires a zero differential there. Transverse intersection eliminates the extra Tor terms when the spectral surface is inserted. Principal translations and the specified normalization supply the global zero-section identity without an unprinted base line. A differently normalized kernel, nonprincipal fibration or nontransverse support requires its own formula; it is not covered by (17.1).

In particular,

$$h^0(X, V) = h^3(X, V) = 0, \quad h^1(X, V) = h^0(\Sigma, M), \quad h^2(X, V) = h^1(\Sigma, M).$$

The Euler characteristic changes sign. This upgrades the earlier unordered localization contract to an exact oriented statement on the declared domain, without asserting that its selected physical inputs have already been supplied.

## 17.5 The ordinary representative fails, but the selected line decides

For the frozen eta9 graph-Prym route, the product Jacobian has  $H^2 = 2$ ,  $D_0^2 = 0$ , and spectral class  $9H + 3D_0$ . Consequently the smooth matter curve has class  $9H$  on the K3,

$$g(\Sigma) = 1 + \frac{(9H)^2}{2} = 82, \quad \deg(H|_\Sigma) = 18.$$

The three graph degrees 3, 3, 6 cut a length-12 cycle  $Z$ ; the ordinary restricted line is

$$M_{\text{alg}} = \mathcal{O}_\Sigma(5H)(-Z), \quad \deg M_{\text{alg}} = 78, \quad \chi(M_{\text{alg}}) = -3.$$

The K3 restriction sequence from  $\mathcal{O}_B(-4H)$  to  $\mathcal{O}_B(5H)$ , with Kodaira/Serre vanishing, gives  $h^0(\Sigma, \mathcal{O}_\Sigma(5H)) = 27$ . Twelve point conditions remove at most twelve sections. Hence

$$h^0(M_{\text{alg}}) \geq 15, \quad h^1(M_{\text{alg}}) \geq 18.$$

Under the localization hypotheses the corresponding component count is at least  $16(15+18) = 528$ , not 48. This rejects the *uncorrected ordinary representative* as a mirror-free source. It does not construct that representative as an admissible twisted endpoint or exclude the completed twisted construction.

The relevant line is instead

$$M_* = M_{\text{alg}} \otimes F_\alpha, \quad F_\alpha \in \text{Pic}^0(\Sigma).$$

Here  $F_\alpha$  compares an explicit cover trivialization  $b_C$ , available after the cover obstruction vanishes, with  $t_0$  on the intersection. Degree zero follows after enforcing the fixed Chern/index rows in the same normalization. Zero obstruction is an existence condition, not the printed cochain or its comparison line. Ambient zero-section normalization does not force  $F_\alpha$  to be trivial.

For smooth ample  $C \subset B \times E^\vee$ , weak Lefschetz identifies  $\text{Pic}^0(C)$  with the elliptic-factor Picard variety. Those lines restrict trivially to  $\Sigma \subset B \times \{0\}$ . Thus the ordinary fixed-Chern ambiguity on the cover is not a knob that freely tunes the matter line. This conclusion uses ampleness; changing that domain would require a fresh restriction calculation.

## 17.6 A finite test, not a substitute matrix

Mirror freedom on this route is the exact condition  $h^0(M_*) = 0$ . The effective locus  $W_{78}^0(\Sigma) \subset \text{Pic}^{78}(\Sigma)$  has dimension at most 78, while the Picard variety has dimension 82. Generic degree-zero twists therefore avoid it, but this says nothing about where the selected  $F_\alpha$  lands. The *theta test* here means this line-cohomology non-speciality test; it is not a weak-angle prediction or a reuse of observed flavor data.

Choose an effective divisor  $A$  of degree 85. Then

$$\deg M_*(A) = 163 > 2g - 2 = 162, \quad h^0(M_*(A)) = 82.$$

The evaluation sequence gives one map

$$\text{Ev}_A : H^0(M_*(A)) \longrightarrow H^0(M_*(A)|_A), \quad \text{matrix size } 85 \times 82.$$

If its rank is  $r$ , exactness gives the entire multiplicity calculation:

$$h^0(M_*) = 82 - r, \quad h^1(M_*) = 85 - r.$$

Full column rank, certified by one nonzero  $82 \times 82$  minor, is therefore equivalent to mirror freedom. A rank deficiency  $m = 82 - r$  gives  $48 + 32m$  components. An identity block padded with three rows shows algebraic feasibility, not the value of this geometric map. Its entries must be evaluations of sections of the same selected line, with bases, divisor and exactness certificate.

For the  $c_3(V) = +6$  normalization, the derived shift fixes

$$\chi(X, V) = 3, \quad (h^1(X, V), h^2(X, V)) = (0, 3)$$

when the test passes. Dualization swaps the middle lanes; the naming of matter and antimatter follows the representation convention. This is not a new selection of a time arrow. Dimensions do not require a metric, but normalized harmonic representatives and their Yukawa overlaps do. The physical exotic/Higgs operator remains a separate same-source export.

The resulting order is now concrete: selected cover and twisted line, normalized restriction and transversality, evaluation-map rank, then harmonic normalization and physical lifting. The localization theorem and finite test are available. A completed obstruction calculation must still emit its trivialization and line before it can instantiate them. This organization preserves the existing finite SM/profile results while identifying what connects them to this geometric source.

## 18 The cohesive closure-repair theorem

The preceding results can now be stated as one composition theorem.

**Theorem 18.1** (Cohesive closure-repair and shadow theorem). *Suppose the following data are supplied:*

1. *a metric configuration space with a smooth defect section  $\Phi$ , smooth metric data and an exact-zero background  $p_*$  (a  $C^N$  source gives only the cost jet through the available order);*
2. *compatible local isometric/unitary descent data;*
3. *on the cohesive branch, an integrable possibly twisted superconnection whose endomorphism algebra realizes the defect complex;*
4. *a Hermitian structure and self-adjoint elliptic domains; and*
5. *any declared orthogonal projection or isometric transform reducing the linearized operator.*

*Then one and the same source determines, functorially:*

$$\begin{aligned} \mathcal{C} &= \frac{1}{2} \|\Phi\|^2, & X &= -\text{grad } \mathcal{C}, & H &= D\Phi(p_*)^\dagger D\Phi(p_*), \\ e^{-sH}, & P_0, & G &= H^{-1}(I - P_0), & \chi_I(H), & K_f(x, y) = \langle x | f(H) | y \rangle, \\ & \{D^n \mathcal{C}(p_*)\}_{n \geq 3} & & \text{and their finite formal repair graphs.} \end{aligned} \quad (18.1)$$

*The nonlinear flow, tangent calculus, and existing basin limits descend through the compatible local chart equivalences. Reducing projections transport the tangent functional calculus. An onto unitary equivalence of the full residual and metric transports every object in (18.1). An embedded nonlinear subsystem additionally needs invariance of its repair vector field; exact retained graph transport requires compatibility of the full typed vertex, grading and contraction data, as in the diagrammatics theorem. Reduction of the quadratic operator alone does not supply these nonlinear hypotheses.*

*The theorem does not identify  $\mathcal{C}$  with a signed physical action. Such an identification requires a variational anchor or a separately typed multiplier/cyclic construction. Nor does the theorem select the original defect, metric, background, causal domain, state, or projection.*

*Proof.* The first variation, full Hessian, and exact-zero normal square are [theorems 2.1](#) and [2.2](#). The repair flow and its tangent semigroup are [theorem 3.1](#). Cohesive realization and twist cancellation are [theorems 4.1](#) and [4.2](#). Elliptic Hodge theory supplies the functional calculus in [theorem 6.1](#). Nonlinear descent and compatible transport are [theorems 7.1](#) and [7.2](#). The complete cost jet is [theorem 13.1](#); formal Wick contraction then produces its graph weights. The final nonidentification statement follows from [theorems 14.1](#) to [14.4](#).  $\square$

The theorem says something stronger than “an operator generates repair.” The nonlinear repair field exists first. Its derivative produces the operator. Yet it also says something more careful than “closure explains all operators.” Only the displayed class of positive tangent operators is obtained, and only after its source data are supplied.

## 19 The q79 benchmark and its exact boundary

### 19.1 What is already constructed

The current q79 research program provides a hash-bound  $\alpha$ -twisted cohesive benchmark  $\mathcal{S}_{\text{HS}}$  on  $J$ . On that object, the following are exact:

1. the twisted cohesive representative and ordinary global endomorphism dg algebra;
2. the canonical Maurer–Cartan curvature residual;
3. the positive Hodge repair Hessian after a Hermitian choice;
4. the tangent heat semigroup, harmonic projector, and reduced Green operator on the global total space;
5. projective local descent of the finite coherent repair witness; and
6. conditional transport of Ext, Yoneda, and formal Maurer–Cartan data through a dg-enhanced BHT/Fourier–Mukai equivalence.

The q79 result is not merely a finite analogy. It applies the general mathematics to one explicit twisted perfect-complex atlas. But it is a benchmark object, not yet the selected physical visible/hidden endpoint.

## 19.2 The proposed full defect and why it remains proposed

The physical source program organizes the Hull–Strominger obligations into a multi-lane residual of the schematic form

$$\Phi_{\text{phys}} = (\Phi_0, \mu_{TX}, \mu_V, \mu_W, \Phi_{\text{bal}}, \Phi_{\text{AB}}, \Phi_{\text{SU}(3)}). \quad (19.1)$$

Here  $\Phi_0$  denotes integrability, the  $\mu$ -rows denote HYM moment maps,  $\Phi_{\text{bal}}$  balanced geometry,  $\Phi_{\text{AB}}$  anomaly/Bianchi closure, and the final row the declared  $\text{SU}(3)$  normalization or coordinate constraint.

If a selected endpoint  $s_*$  satisfies  $\Phi_{\text{phys}}(s_*) = 0$ , then the positive physical repair Hessian is exactly

$$A_* = D\Phi_{\text{phys}}(s_*)^\dagger D\Phi_{\text{phys}}(s_*). \quad (19.2)$$

Writing the linearization as a deformation-complex block  $J_*$  and additional physical rows  $K_*$  gives

$$A_* = J_*^\dagger J_* + K_*^\dagger K_*. \quad (19.3)$$

This is an exact identity *if the same selected endpoint, metric, and domains supply every row*. The current corpus has not yet supplied that complete physical object. Equation (19.3) is therefore the correct target theorem, not a declaration that the target has already been reached.

## 19.3 The explicit residual and its block support

The full-residual record improves the schematic notation in (19.1) to an explicit differential-operator compiler. Write  $\nu = \|\Omega_{\text{hol}}\|_\omega$ , and reserve  $c_{\text{vol}}$  for the volume normalization constant, not a third Chern class. Let  $\Psi$  denote the independently varied volume-form representative in the ambient compiler; it is not silently identified with  $\Omega_{\text{hol}}$ . The six extra rows are

$$\begin{aligned} \mu_{TX} &= R_\Theta \wedge \omega^2, & \mu_V &= F_V \wedge \omega^2, & \mu_W &= F_W \wedge \omega^2, \\ \mathcal{B} &= d(\nu\omega^2), \\ \mathcal{A} &= dH - \frac{\alpha'}{4}(\text{tr } R_\Theta^2 - \text{tr } F_V^2 - \text{tr } F_W^2), \\ \mathcal{N} &= i\Psi \wedge \bar{\Psi} - c_{\text{vol}}\omega^3. \end{aligned} \quad (19.4)$$

Their target is the orthogonal sum of the six spaces

$$\Omega^6(\text{ad } TX), \quad \Omega^6(\text{ad } V_3), \quad \Omega^6(\text{ad } W_9^\tau), \\ \Omega^5, \quad \Omega^4, \quad \Omega^6.$$

This is a choice of positive repair metric after a common scale is factored out. It is not an identification with a Lorentzian action.

At a fixed complex-structure chart, put  $a = (a_{TX}, a_V, a_W)$ . Exterior differentiation and the covariant curvature variation give

$$\begin{aligned} K_{TX}v &= d_\Theta a_{TX} \wedge \omega^2 + 2R_\Theta \wedge \omega \wedge \dot{\omega}, \\ K_Vv &= d_{A_V} a_V \wedge \omega^2 + 2F_V \wedge \omega \wedge \dot{\omega}, \\ K_Wv &= d_{A_W} a_W \wedge \omega^2 + 2F_W \wedge \omega \wedge \dot{\omega}, \\ K_Bv &= d\{\nu(2\omega \wedge \dot{\omega} + \log \dot{\nu} \omega^2)\}, \\ K_{\mathcal{A}}v &= d\dot{H} - \frac{\alpha'}{2}\{\text{tr}(R_\Theta \wedge d_\Theta a_{TX}) - \text{tr}(F_V \wedge d_{A_V} a_V) - \text{tr}(F_W \wedge d_{A_W} a_W)\}, \\ K_{\mathcal{N}}v &= i(\dot{\Psi} \wedge \bar{\Psi} + \Psi \wedge \dot{\bar{\Psi}}) - 3c_{\text{vol}}\omega^2 \wedge \dot{\omega}. \end{aligned} \tag{19.5}$$

Here  $\log \dot{\nu} = \text{Re}\langle \dot{\Omega}_{\text{hol}}, \Omega_{\text{hol}} \rangle / \|\Omega_{\text{hol}}\|^2 - \frac{1}{2} \text{tr}_g \dot{g}$  in the fixed-chart metric convention. Varying complex structure requires the corresponding typed chart terms as well; these formulas do not supply their endpoint coefficients. In particular the factor  $\alpha'/2$ , rather than  $\alpha'/4$ , is forced by differentiating the curvature square. On the normalized chart  $\Psi = \Omega_{\text{unit}}$ , with matching volume constants,  $\mathcal{N}$  is an identity and its tangent derivative vanishes. If instead  $\Psi = \Omega_{\text{hol}}$ , its vanishing imposes the additional constant-norm condition discussed in [section 15.3](#). This role clarification preserves the six-row ambient differentiation, while avoiding an unintended restriction on conformally balanced solutions.

The rank-102 complex fiber is

$$\mathcal{Q} = T^*X \oplus \text{ad}(TX) \oplus \text{ad}(V_3) \oplus \text{ad}(W_9^\tau) \oplus TX, \quad \mathbf{r} = (3, 8, 8, 80, 3).$$

It is not the full form-augmented Hilbert space and not a finite discretization dimension. In this lane order the extra-row incidence is

$$I_K = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

A one means allowed dependence, not a certified nonzero coefficient. The anomaly row has a common target for all five lanes. Its Gram square therefore allows all 25 ordered blocks and  $102^2 = 10\,404$  ordered fiber positions. Omitting it had allowed only 19 blocks and 7716 positions. The six cross-gauge blocks contribute  $2(8 \cdot 8 + 8 \cdot 80 + 8 \cdot 80) = 2688$  positions. They must not be forced to zero before evaluating the same-source operator. Nor may the real Bianchi row be deleted as already contained in the complex without a same-domain isometry proving that exact redundancy.

This correction changes the possible coupling pattern, not the positivity identity. On the common closed form domain,

$$H_{\text{phys}} = \Delta_{\mathcal{Y},1} + K^\dagger K \geq \Delta_{\mathcal{Y},1}, \quad \ker H_{\text{phys}} = \ker \Delta_{\mathcal{Y},1} \cap \ker K.$$

The finite witness in the source has  $\Delta = \text{diag}(0, 0, 1, 2, 4)$  and  $K = (1, 1, 0, 0, 0)$ . It removes one of two harmonic directions and retains a positive gap of one. That example shows why neither the harmonic projector nor a scalar rescaling of the bare Hodge operator can be assumed.

There is also a lawful conditional sparsification. If the physical endpoint, residual, metric and Galerkin projection all respect the projective  $E[3]$  action, character orthogonality removes unequal-character blocks. The hidden adjoint multiplicities are 8 for the trivial character and 9 for each of eight others. If, additionally, all 22 non-hidden modes are trivial, the count becomes

$$30^2 + 8 \cdot 9^2 = 1\,548.$$

The general count is  $\sum_{\chi} (m_{\text{ext},\chi} + m_{\text{hid},\chi})^2$ . The value 1 548 is not a selected physical matrix count until these representation and invariance hypotheses are verified. It complements, and does not retract, the unconditioned 25-block support.

These are imported compiler results, not newly proved source existence. The earlier candidate audit in that record is historical. In particular, later hidden-carrier and existential HYM progress retained in the Flux paper is not undone by its old status fields. A common physical visible endpoint, all differential residual rows and their numerical coefficients are the stronger input needed here.

#### 19.4 The shared circle

The shared circle enters this construction as a differential line  $(L_{\text{sh}}, \nabla_{\text{sh}})$  and its equivariance, not as an extra coordinate and not as physical time. The finite q79 theorem already shows that one universal flat cyclic line acts compatibly on the selected finite carriers and commutes with their projectors and finite Hessian. The continuum promotion must prove the corresponding relation

$$[\nabla_{\text{sh}}, D\Phi_{\text{phys}}] = 0$$

in the appropriate equivariant sense. It would then follow that the repair Hessian and all its functional-calculus shadows preserve the shared charge sectors. This is a serious unification condition. It does not identify compact phase with Lorentzian time.

## 19.5 Status table

| Object or arrow                                                       | Status                     | Boundary                                                         |
|-----------------------------------------------------------------------|----------------------------|------------------------------------------------------------------|
| General covariant residual Hessian                                    | Exact here                 | Includes nonzero-defect correction                               |
| Exact-zero normal square and repair semigroup                         | Exact here                 | Requires selected metric and domain                              |
| Twisted cohesive endomorphism dg algebra on $\mathcal{S}_{\text{HS}}$ | Exact structural benchmark | Scalar twist cancels in End                                      |
| Maurer–Cartan repair and Hodge tangent on $\mathcal{S}_{\text{HS}}$   | Exact structural benchmark | Hermitian metric exists but is not physically selected           |
| Local nonlinear repair descent                                        | Exact here                 | Does not select one basin                                        |
| Isometric functional-calculus transport                               | Exact conditional          | Arbitrary derived equivalence is not unitary                     |
| Repair jet to formal graph data                                       | Exact finite/formal        | Not a physical QFT action                                        |
| Cyclic action for the integrability lane                              | Exact conditional          | Requires cyclic trace and domain hypotheses                      |
| Anomaly triple to string algebroid                                    | Exact constructor          | Supplied holomorphic bundles, pairing and anomaly data           |
| q79 Bott–Chern MC lift                                                | Exact on stated sublocus   | Nonlinear section, not a full cohesive equivalence               |
| Calabi role reduction                                                 | Exact conditional          | Matched conventions; pseudo-Kahler is not automatically positive |
| Selected physical $V_3/W_9$ cohesive or pure-bundle endpoint          | Open                       | Current source-selection frontier                                |
| Full HYM/balanced/anomaly action and real slice                       | Open                       | Current action frontier                                          |
| Lorentzian causal operator and state                                  | Open                       | Cannot be inferred from the positive square                      |
| Continuum-to-finite product/Hodge intertwiner                         | Open                       | Current geometry frontier                                        |

## 20 What the theorem changes downstream

### 20.1 Fixed points

The fixed-point theory keeps its existing theorem ownership. It proves existence, uniqueness under suitable hypotheses, smoothing, projection control, and convergence. The present paper supplies one principled source for the energy and tangent generator to which those theorems may be applied. It does not repeat their analytic hypotheses or promote a finite witness to an infinite-dimensional flow.

### 20.2 Wave and particle representations

The positive filtered operator in the wave–particle paper no longer needs to be viewed as mathematically arbitrary on the cohesive benchmark. It can be the functional calculus of a repair Hessian. Its local and spectral descriptions then become the localized and normal-mode shadows of one tangent repair process. The quantum state, Born bridge, instrument, and physical time evolution remain separate obligations.

### 20.3 Perturbative graphs

The graph paper begins with typed quadratic and higher tensors. The repair jet theorem derives one entire such tensor family from  $(\Phi, W)$ . This removes independent repair vertices as source inputs. It does not remove physical couplings from a QFT until the repair functional is proved to be the physical signed action or is related to it by a certified variational map.

### 20.4 Quantum field theory

The existing selected free q79 CAR/AQFT result uses standard quantum-field machinery after its geometric input is supplied. The present paper does not rederive that machinery. It clarifies the upstream division of labor: positive repair controls stabilization and Euclidean/Hodge localization; the signed gauge-fixed action controls physical response and causal Green operators; the state and renormalization prescription complete the quantum theory.

### 20.5 Gravity

If a gravitational defect system is selected, its positive repair Hessian and kernel follow by the same theorem. This supports the MTT intuition that geometry may respond to closure pressure. It does not prove that Einstein's equations are the selected repair equation, that the stress tensor has the required normalization, or that the resulting theory is UV complete. The Lorentzian principal symbol and causal domains must be derived from the signed physical system, not guessed from a positive Hilbert-space metric.

## 21 Theorem ownership and corpus placement

This paper is intended to prevent the same upstream theorem from being reproved in several downstream papers.

| Result                                                  | Canonical owner                                                | Use elsewhere                                       |
|---------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------|
| Covariant residual Hessian and exact-zero normal square | This paper                                                     | Foundations, Fixed Points, QG may cite              |
| Repair flow to tangent heat semigroup                   | This paper                                                     | Wave-particle and stabilization papers may cite     |
| Cohesive q79 realization and twist cancellation         | This paper as synthesis; executable theorem packet as evidence | Source and bundle papers may cite                   |
| Repair descent and compatible functional calculus       | This paper                                                     | Projection papers may import conclusion             |
| Repair jet to formal graph tensors                      | This paper                                                     | Modal diagrammatics retains graph theorem ownership |
| Finite graded Wick and graph-projection theorem         | Modal diagrammatics                                            | Imported here without proof duplication             |
| Fixed-point existence and convergence                   | Fixed Points I and successors                                  | Imported here                                       |
| Conditional wave-particle operator encoding             | Wave-particle paper                                            | Receives a structural source candidate here         |
| Physical q79 endpoint and action                        | No completed owner yet                                         | Research frontier, not a theorem in this paper      |

The new result should therefore be added to the related papers by a concise cross-reference and status update, not by copying the proofs. This keeps the corpus readable and makes later corrections local.

## 22 Falsifiers and acceptance tests

The structural theorem is falsified in an application if any required map or domain fails. The physical interpretation fails if a stronger claim is made without its bridge. A candidate promotion should pass the following tests.

1. **Defect source:** specify every residual row, bundle, connection, domain, and source hash.
2. **Background:** prove whether the selected state has zero defect. If it does not, compute the residual Hessian correction.
3. **Metric:** derive or explicitly declare the field and residual metrics; do not infer them from a categorical equivalence.
4. **Descent:** verify the literal overlap maps, cocycles, and isometric/unitary compatibility.
5. **Hodge domain:** prove self-adjointness, ellipticity or the appropriate hyperbolic replacement, and boundary conditions.
6. **Projection:** verify reduction of the operator and preservation of the nonlinear residual jet, not only matching dimensions.
7. **Action:** provide a Helmholtz/cyclic anchor or a fully typed multiplier/BV lift. Do not call the positive square the physical action.
8. **Causality:** derive retarded, advanced, or Feynman prescriptions from the signed gauge-fixed operator and state.
9. **Strata:** test Green and contraction bounds near every cohomology-rank jump.
10. **Numerics:** evaluate endpoint coefficients only after the same source has passed the structural tests.

These tests turn the paper into a research instrument. A failure identifies which arrow is missing; it does not send the program back to the beginning.

## 23 Frontier theorem

The later endpoint-factorization record organizes seven acceptance rows into three structured sources:

1. **GAS**, geometry and action: the physical endpoint and signed upper action;
2. **SYN**, spectral synthesis: the analytic intertwiner and compatible chain/product contraction data;
3. **BV4**, a BV-compatible four-dimensional compactification.

Shared-line functional calculus and the Galerkin/Feshbach execution are then consequences of compatible GAS and SYN data, rather than two more independent source objects. Three structured sources are not three scalar parameters. None of the three complete physical packages is accepted in the cited record. This dependency reduction preserves the exact finite constructions while preventing another round of inventing independent kernels for already derived arrows.

The decisive next theorem is not another abstract Hessian statement. That part is now complete. It is a selected-source theorem of the following form.

**Conjecture 23.1** (Selected physical cohesive-source theorem). There exists a selected q79 Hull–Strominger endpoint  $s_*$ , represented either by a stable pure-bundle source or by a physically admitted cohesive source, together with:

1. the full defect section  $\Phi_{\text{phys}}$  and a common Hermitian metric;
2. the shared differential line and equivariant connections;
3. a cyclic or BV variational structure whose Euler–Lagrange rows agree with the selected physical residual on a declared gauge slice;
4. Lorentzian externalization, real structure, and causal domains; and
5. a continuum-to-finite chain, product, Hodge, and normalization intertwiner with an error certificate.

At an exact endpoint, the positive repair operator is  $A_* = D\Phi_{\text{phys}}^\dagger D\Phi_{\text{phys}}$ , while the signed physical operator is obtained from the certified variational structure rather than inferred from  $A_*$ .

Proving this conjecture would connect the upper bundle geometry, cohesive deformation theory, shared phase, fixed points, wave kernels, finite q79 operators, and the signed physical theory without conflating them. It is a sharp target because every required object has a declared type and every invalid shortcut has already been removed.

## 24 Discussion

### 24.1 What has genuinely been gained

The main gain is upstream organization. The heat kernel is no longer merely an attractive analogy for coherence. Under the theorem’s hypotheses it is the exact tangent propagator of nonlinear defect repair. The harmonic sector is the kernel of the same Hessian. The spectral projector and local kernel come from its functional calculus. The formal repair vertices are higher derivatives of the same cost. Compatible projection preserves the whole package.

This makes several previously separate papers part of one coherent story:

$$\begin{aligned}
 &\text{closure defect and metric} \longrightarrow \text{repair flow}, \\
 &\text{repair flow} \longrightarrow \text{Hodge Hessian}, \\
 &\text{Hodge Hessian} \longrightarrow \text{heat, Green, and spectral kernels}, \\
 &\text{higher repair jet} \longrightarrow \text{formal graph grammar}, \\
 &\text{compatible projection} \longrightarrow \text{retained shadows}.
 \end{aligned}$$

### 24.2 What has not been gained

The same clarity exposes the missing physics. A positive normal square cannot tell us which first-order sign or phase nature uses. It cannot choose a Lorentzian causal prescription. It cannot distinguish a physical saddle from a repair minimum. It cannot supply a quantum state, a Born rule, or a renormalization scheme. These are not small numerical corrections; they are different typed structures.

The result is therefore neither “only mathematics” nor a completed theory of everything. It is a rigorous mathematical bridge that narrows the physical frontier. It tells us which

downstream constructions are already forced once a selected source exists, and which ones still require genuinely new input.

### 24.3 Why the cohesive route matters

The cohesive route is especially useful because it remains regular where an open-stratum bundle contraction fails. It allows the full derived object to carry a global differential, deformation algebra, and Hodge calculus without first pretending that the cohomology rank is constant. This may be the right preprojection language for MTT. It may also turn out to be only an auxiliary resolution of a physical pure-bundle endpoint. The present theorem permits both outcomes and gives a concrete test between them.

## 25 Conclusion

Closure is not represented here first as a static operator. It is represented by a nonlinear defect and the process that reduces its cost. The operator appears as the derivative of that process at an admissible state. Hodge theory, heat flow, Green response, spectral projection, local kernels, and formal repair graphs then follow as exact shadows of one source.

The result is strongest precisely because its boundary is explicit. At a nonzero-defect critical point the Hessian contains a residual correction. A derived equivalence is not automatically isometric. A global cohesive operator does not guarantee a uniform fiberwise Green family. A positive repair functional is not a signed action. A kernel is not automatically a quantum propagator. A descended basin map is not automatically unique actualization.

For q79, the cohesive Hartshorne–Serre benchmark now realizes the structural chain exactly after a Hermitian choice. The next leap is no longer to guess another operator. It is to select the physical endpoint, metric, cyclic or BV action, Lorentzian domains, and continuum-to-finite intertwiner from one source. If that is achieved, much of the downstream operator language will already be waiting as theorem rather than assumption.

## Computational Evidence and Reproducibility

The universal proofs in this paper are analytic or finite algebraic. Their MTT-specific instantiations are pinned in the curated [MTT Results Reproduction repository](#). The principal result records are:

- the q79 cohesive-superconnection record;
- the q79 Maurer–Cartan repair record;
- the local Čech–Hilbert and nonlinear repair-descent records;
- the covariant residual-Hessian record and nonzero-defect witness;
- the reversible–dissipative repair record;
- the repair-jet graph-functor record;
- the variational-anchor and multiplier-lift record;
- the cyclic Maurer–Cartan action-descent record; and
- the q79 string cyclic-cotangent boundary record.

Version 3 additionally imports:

- the explicit six-row/25-block compiler: `q79_v3w9_full_residual`;

- its conditional character reduction: `q79_rank102_qutrit_reduction`;
- the bounded transported-metric record: `preprojection_bht_transporting_metric`;
- the endpoint factorization record, listed under `causal_base_q79_seven_row_endpoint_factorization_packet`.

Version 4 reviews `q79_continuum_finite_intertwiner_cutset`. Its exact intertwining conditions and rational witnesses are retained; its unqualified whole-projector gap estimate is corrected explicitly in the projection section. The frozen artifact remains unchanged for audit. The local Cech equalizer, nonlinear descent, nonzero-defect example and phase-loss and damping witnesses are explicitly compared in the current contextual integration ledger. These imports use the frozen result manifest at [f141a20ea23c](#); the integration companion gives each artifact hash and manuscript location. Each record declares its theorem tier, source artifacts, verifier, zero-fit status, and nonpromotion guards. The exact witnesses use rational or symbolic arithmetic where stated. They do not select the physical q79 endpoint or supply measured values. The source TeX, generated Markdown, compiled PDF, metadata hashes, and version-provenance audit accompany this paper in the public MTT papers repository. The evidence snapshot used for Version 1 is commit `ba0874012246bb930677ebfa6f0a6eb33a57a8f0`.

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